

Using facet for Generalizability Theory

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Introduction to Generalizability Theory

Generalizability Theory (G-theory) is a comprehensive framework for understanding the reliability of measurements. Developed by Cronbach and colleagues (Cronbach et al., 1963, 1972; Cronbach, Gleser, Nanda, & Rajaratnam, 1972), G-theory extends Classical Test Theory (CTT) by recognizing that measurement error is not monolithic but arises from multiple sources. This allows researchers to examine how different facets of a measurement design—such as items, raters, tasks, or occasions—contribute to measurement error, and to optimize future measurement procedures accordingly.

The **facet** package provides comprehensive tools for conducting G-theory analyses using variance component models. It supports three estimation backends: frequentist approaches using **lme4** (Restricted Maximum Likelihood) and **mom** (Method of Moments/ANOVA-based), and Bayesian estimation using **brms** (No-U-Turn Sampler/Hamiltonian Monte Carlo via Stan). The package provides a unified interface for univariate and multivariate designs, with full support for G-studies, D-studies, and coefficient calculations with uncertainty quantification.

Historical Context and Foundations

Foundational Works in Generalizability Theory

The theoretical foundations of generalizability theory were established through a series of landmark publications:

Cronbach, Rajaratnam, & Gleser (1963) introduced the core concept of “universes of admissible observations” and established the distinction between relative and absolute error, providing the theoretical framework that unified various approaches to reliability estimation. This paper liberated reliability theory from the constraints of parallel forms, recognizing that measurement could be generalized across multiple facets of observation.

Cronbach, Gleser, Nanda, & Rajaratnam (1972) provided the definitive treatment of G-theory in their seminal book *The Dependability of Behavioral Measurements*. This work extended G-theory to multivariate designs, introducing variance-covariance matrix estimation for multiple dimensions and establishing the mathematical framework for profile reliability and composite score generalizability.

Rajaratnam, Cronbach, & Gleser (1965) extended the framework to stratified-parallel tests, developing the nested facet notation (e.g., items nested within subtests) that remains standard in G-theory notation today.

Brennan (2001) synthesized these developments into *Generalizability Theory*, the modern comprehensive treatment that covers unbalanced designs, multivariate extensions, and computational methods. This text remains the definitive reference for G-theory methodology.

Shavelson & Webb (1991, 2006) made G-theory accessible to applied researchers through their primer *Generalizability Theory: A Primer*, which has introduced generations of researchers to G-theory concepts through clear explanations and practical examples.

Computational History and Software Development

The computational implementation of G-theory has evolved through several generations:

Early Methods (1960s-1970s): Initial G-theory analyses relied on hand calculations or general-purpose ANOVA programs (BMDP, SPSS, SAS), limiting applications to balanced designs. Researchers had to compute variance components manually from expected mean squares.

GENOVA (1983): Brennan’s GENOVA program was the first dedicated G-theory software, implementing ANOVA-based variance component estimation for crossed and nested designs. This made G-theory practical for the first time, though it required balanced data.

urGENOVA and mGENOVA (2001): Brennan extended the software suite to handle unbalanced designs (urGENOVA) and multivariate analyses (mGENOVA), significantly expanding the scope of practical G-theory applications.

William P. Vispoel made seminal contributions to G-theory accessibility through his comparative software reviews (Vispoel, 1998, 2000), evaluating programs like GENOVA and providing practical guidance for researchers selecting appropriate computational tools. Vispoel also advanced multivariate G-theory applications in personality assessment (Vispoel & Cook, 1994), demonstrating how G-theory can separate multiple sources of variance in multi-trait, multi-method designs.

G_String provides a modern graphical interface built around urGENOVA, making G-theory accessible without command-line expertise (Bloch & Norman, 2012). This Java-based tool has been particularly valuable in medical education contexts.

Modern R Packages: Contemporary implementations integrate G-theory with mixed-effects modeling frameworks. The **facet** package continues this tradition by providing an R implementation with three estimation backends (ANOVA/MOM, REML, Bayesian) and full support for univariate and multivariate designs, leveraging the computational efficiency of modern tools like lme4 and Stan.

From Classical Test Theory to Generalizability Theory

Classical Test Theory decomposes an observed score X into a true score T and measurement error e :

$$X = T + e$$

The reliability coefficient in CTT, typically Cronbach’s alpha, represents the proportion of observed score variance attributable to true score variance. However, this approach treats all measurement error as undifferentiated—it cannot distinguish between error due to different sources (e.g., items vs. raters) or inform decisions about how to improve measurement precision.

G-theory addresses these limitations by partitioning observed score variance into components attributable to different facets of measurement. Instead of a single reliability coefficient, G-theory provides:

1. **Variance components** quantifying the contribution of each facet to total variance
2. **Generalizability (G) coefficients** for relative decisions (rank-ordering objects)
3. **Dependability (Phi) coefficients** for absolute decisions (comparing to standards)

Key Concepts

Facets are the sources of measurement error in a design. Common facets include:

- **Items:** Questions or tasks administered
- **Raters:** Judges or observers scoring responses
- **Tasks:** Different performance activities
- **Occasions:** Times of measurement
- **Forms:** Parallel versions of an instrument

The **object of measurement** is what we seek to measure—typically persons (individuals, students, patients, etc.). The distinction between facets and the object is crucial: facets are sources of error we seek to control, while the object is the target of inference.

The **universe of admissible observations** defines the set of all acceptable measurement conditions. A G-study estimates variance components from a sample of observations; a D-study uses these estimates to project reliability for different measurement designs within the universe.

Variance components are estimated from a mixed-effects model. For a simple person $\times$ item design:

$$X_{pi} = \mu + \mu_p + \mu_i + \mu_{pi} + e_{pi}$$

Where:

- μ is the grand mean
- μ_p is the effect of person p
- μ_i is the effect of item i
- μ_{pi} is the person $\times$ item interaction
- e_{pi} is residual error

Each effect has an associated variance component: σ_p^2 , σ_i^2 , σ_{pi}^2 , and σ_e^2 .

G-studies (generalizability studies) estimate variance components from observed data. The goal is to understand how much variance is attributable to each facet.

D-studies (decision studies) use G-study variance components to project reliability for different measurement designs. By varying the number of levels of each facet (e.g., number of items, raters), researchers can identify designs that achieve desired reliability levels efficiently.

Univariate Generalizability Theory

Model Specification

For a univariate design with person crossed with item:

$$X_{pi} = \mu + \tau_p + \alpha_i + (\tau\alpha)_{pi} + e_{pi}$$

The variance components are:

- σ_τ^2 (person variance): Universe score variance, reflecting true differences
- σ_α^2 (item variance): Variance due to differences in item difficulty
- $\sigma_{\tau\alpha}^2$ (person $\times$ item): Interaction variance
- σ_e^2 (residual): Unexplained variance

Specifying All Possible Variance Components

A well-specified G-study model contains **every main-effect random factor** and **every estimable interaction** between those factors—no more and no less. This is the rule we follow throughout the **facet** package and the rule that every example in this vignette obeys. Two practical subtleties determine which interactions are estimable in a given dataset:

1. **Interactions confounded with the residual must be omitted.** When an interaction has as many unique combinations as observations (i.e., one observation per cell), the interaction is algebraically indistinguishable from the residual error. The **lme4** backend will refuse to fit such a model, and the **mom** and **brms** backends will silently absorb that interaction into the residual. The correct specification is to leave it out of the formula. In every built-in dataset that has one observation per fully crossed cell, the highest-order interaction is therefore omitted from the canonical formula.
2. **Nested random effects can be written either way when labels are unique.** If facet *A* is nested in facet *B* and the *A* labels are already unique within each level of *B* (e.g., “Item 1” in Subtest 1 is a different item from “Item 1” in Subtest 2), then $(1 \mid A)$ and $(1 \mid A:B)$ produce the same fitted model. We use the un-nested form in canonical formulas because it makes the “all main effects + all estimable interactions” structure more transparent and lets **dstudy()** refer to the facet by a single, unbackticked name.

Worked example 1 (brennan data). The **brennan** data is a Person $\times$ Task $\times$ (Rater:Task) design with one observation per Person $\times$ Rater:Task cell. The canonical formula is:

```
g <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +  
             (1 | Person:Task),  
            data = brennan)
```

Each term is interpretable: **Person** is the universe-score variance, **Task** is task-difficulty variance, **Rater** is within-task rater-lenience variance (the 12 **Rater** labels are unique to each task, so $(1 \mid \text{Rater})$ and $(1 \mid \text{Rater:Task})$ are equivalent), and **Person:Task** is the person-by-task interaction. The Person $\times$ Rater:Task interaction is **not** in the formula because each Person $\times$ Rater:Task cell has a single observation, so that effect is confounded with the residual and is absorbed into the error term.

Worked example 2 (rajaratnam data). The **rajaratnam** data is a Person $\times$ Subtest $\times$ (Item:Subtest) design with one observation per Person $\times$ Item:Subtest cell. The dataset ships with both an **Item** column (the original 4-level Rajaratnam et al. (1965) factor, whose labels are reused across subtests) and an **ItemId** column (an 8-level disambiguated factor, formed as `paste(Subtest, Item, sep = ".")`, e.g. “1.1”, “1.2”,

“2.1”, “2.2”, “2.3”, “2.4”, “3.1”, “3.2”). The canonical formula uses `ItemId` so the within-subtest nesting of items is visible in the formula:

```
g <- gstudy(Score ~ (1 | Person) + (1 | Subtest) + (1 | ItemId) +
            (1 | Person:Subtest),
            data = rajaratnam)
```

Because `ItemId` is unique within `Subtest`, `(1 | ItemId)` and `(1 | Item:Subtest)` are exactly equivalent random-effect specifications. The `Person × Item:Subtest` interaction is **not** in the formula because each `Person × Item:Subtest` cell has a single observation, so that effect is confounded with the residual and is absorbed into the error term.

Worked example 3 (fully crossed `Person × Task × Rater`). If every rater rates every person on every task (one observation per cell), the canonical formula is:

```
g <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
            (1 | Person:Task) + (1 | Person:Rater) + (1 | Task:Rater),
            data = fully_crossed_data)
```

All three main effects and all three two-way interactions are estimable; the three-way `Person:Task:Rater` interaction is confounded with the residual and is therefore omitted.

Why this rule matters. Under-specifying a G-study (omitting an estimable interaction) is a form of model mis-specification: the omitted variance is silently absorbed by the included terms, biasing the variance-component estimates and the G/Phi coefficients computed from them. Cronbach, Gleser, Nanda, & Rajaratnam (1972) and Brennan (2001) are emphatic on this point—a properly specified G-theory model includes every estimable interaction by design. Over-specifying (adding a confounded interaction) is detected and rejected by `lme4`; with `mom` or `brms` it is silently absorbed into the residual, which is harmless but obscures the model structure. The canonical formulas in this vignette always take the middle path: every estimable interaction is present, and only those.

Relative and Absolute Error

G-theory distinguishes between two types of measurement error:

Relative error (δ) affects rank-ordering of persons. For a crossed design:

$$\sigma_{\delta}^2 = \frac{\sigma_{\tau\alpha}^2}{n_i} + \frac{\sigma_e^2}{n_i}$$

Relative error includes components that involve the object of measurement—variance that causes persons’ scores to shift relative to each other.

Absolute error (Δ) affects the absolute magnitude of scores:

$$\sigma_{\Delta}^2 = \frac{\sigma_{\alpha}^2}{n_i} + \frac{\sigma_{\tau\alpha}^2}{n_i} + \frac{\sigma_e^2}{n_i}$$

Absolute error includes all facets except the object, because any systematic facet effect shifts the observed mean away from the true mean.

Reliability Coefficients

The **generalizability coefficient** (G coefficient, $E\rho^2$) quantifies reliability for relative decisions:

$$E\rho^2 = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma_\delta^2}$$

The **dependability coefficient** (Phi coefficient, Φ) quantifies reliability for absolute decisions:

$$\Phi = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma_\Delta^2}$$

Both coefficients range from 0 to 1, with higher values indicating greater reliability. The Phi coefficient is always less than or equal to the G coefficient because absolute error includes more sources of variance.

Crossed and Nested Designs

In a **crossed design**, all levels of one facet appear with all levels of another. A person $\times$ item crossed design ($p \times i$) means all persons respond to all items.

In a **nested design**, levels of one facet appear with only one level of another. If items are nested within subtests ($i : s$), each item appears in only one subtest.

The variance component structure differs for nested designs. For items nested in subtests:

$$X_{psi} = \mu + \tau_p + \sigma_s + \iota_{i:s} + (\tau\sigma)_{ps} + (\tau\iota)_{pi:s} + e_{psi}$$

The nesting relationship affects how variance components are interpreted and how they contribute to error variance.

Multivariate Generalizability Theory

Historical Development of Multivariate G-Theory

The extension of G-theory to multiple dimensions was first developed in **Cronbach, Gleser, Nanda, & Rajaratnam (1972)**, which introduced variance-covariance matrices for random effects across dimensions. This foundational work established the mathematical framework for profile reliability and composite score generalizability.

Webb & Shavelson (1981) demonstrated practical applications of multivariate G-theory to educational assessment in a seminal paper that showed how multiple correlated dependent variables could be analyzed simultaneously. This work made multivariate G-theory accessible to applied researchers.

Joe & Woodward (1976) established theoretical connections between multivariate G-theory and canonical correlation analysis, demonstrating that multivariate generalizability equals squared canonical correlation under certain conditions.

Marcoulides (1990, 1994, 1996) made significant contributions to multivariate G-theory methodology, developing alternative estimation approaches including SEM-based methods and examining weighting schemes for composite reliability. His work addressed the critical question of how to weight dimensions when forming composites.

Brennan, Kim, & Lee (2022) extended multivariate G-theory to handle complex design structures where different random-effects structures exist for each level of a fixed facet, addressing limitations in traditional multivariate approaches.

Jiang, Raymond, Shi, & DiStefano (2020) provided modern R implementation for multivariate G-theory using linear mixed-effects model frameworks, bridging G-theory with contemporary mixed model methodology.

Vispoel, Lee, Hong, & Chen (2023) provided a comprehensive modern treatment of multivariate G-theory for psychological assessment contexts, demonstrating applications to personality measures and providing worked examples with real data.

Model Specification

For a multivariate design with D dimensions, the model becomes:

$$\mathbf{X}_{pi} = \mu + \tau_p + \alpha_i + (\tau\alpha)_{pi} + \mathbf{e}_{pi}$$

Where bold notation indicates vectors of length D (one element per dimension).

The key extension is that variance components become **variance-covariance matrices**:

$$\Sigma_\tau = \begin{pmatrix} \sigma_{\tau_1}^2 & \sigma_{\tau_1\tau_2} \\ \sigma_{\tau_1\tau_2} & \sigma_{\tau_2}^2 \end{pmatrix}$$

The off-diagonal elements represent **covariances** between dimensions, quantifying the extent to which random effects are correlated across dimensions.

Correlated Random Effects

When random effects are correlated across dimensions, we can estimate:

- **Within-dimension variance:** $\sigma_{\tau_d}^2$ for dimension d
- **Between-dimension covariance:** $\sigma_{\tau_d\tau_{d'}}$ between dimensions d and d'
- **Between-dimension correlation:** $\rho_{\tau_d\tau_{d'}} = \frac{\sigma_{\tau_d\tau_{d'}}}{\sqrt{\sigma_{\tau_d}^2\sigma_{\tau_{d'}}^2}}$

High correlations indicate that persons' rankings are similar across dimensions, suggesting that dimensions measure related constructs or share common method variance.

Multivariate D-Studies

For each dimension d , reliability coefficients are computed separately:

$$E\rho_d^2 = \frac{\sigma_{\tau_d}^2}{\sigma_{\tau_d}^2 + \sigma_{\delta_d}^2}$$

$$\Phi_d = \frac{\sigma_{\tau_d}^2}{\sigma_{\tau_d}^2 + \sigma_{\Delta_d}^2}$$

The error variances incorporate the dimension-specific variance components.

Wide-Format vs Long-Format Models

Wide-format models assume all dimensions have the same sample size (balanced design). Each dimension is a separate column:

```
# Wide format: Task1, Task2, Task3 as separate columns
gstudy(mvbind(Task1, Task2, Task3) ~ (1|Person) + (1|Rater), data = data_wide)
```

Long-format models allow different sample sizes per dimension (unbalanced designs). A single score column with a dimension indicator:

```
# Long format: single Score column with Subtest indicator
gstudy(bf(Score ~ 0 + Subtest + (0+Subtest|r|Person), sigma ~ 0 + Subtest),
      data = data_long, backend = "brms",
      iter = 2000, cores = 4, refresh = 1000)
```

Long-format models are more flexible but require Bayesian estimation.

Estimation Methods

G-theory variance components can be estimated using several methods, each with distinct advantages and limitations. The **facet** package supports three estimation approaches: Method of Moments (ANOVA-based), Restricted Maximum Likelihood (via lme4), and Bayesian estimation (via brms/Stán).

Expected Mean Squares

Understanding variance component estimation requires understanding Expected Mean Squares (EMS). For any ANOVA source, the EMS expresses what that mean square estimates in terms of population variance components. The Method of Moments, REML, and other estimation approaches all build on this framework.

Deriving EMS: Person $\times$ Item Design

Consider a balanced design where n_p persons each respond to n_i items. The ANOVA table is:

Source	df	SS	MS	EMS
Person (p)	$n_p - 1$	SS_p	MS_p	$\sigma_e^2 + n_i\sigma_{pi}^2 + n_i\sigma_p^2$
Item (i)	$n_i - 1$	SS_i	MS_i	$\sigma_e^2 + n_p\sigma_{pi}^2 + n_p\sigma_i^2$
$p \times i$	$(n_p - 1)(n_i - 1)$	SS_{pi}	MS_{pi}	$\sigma_e^2 + \sigma_{pi}^2$

Derivation of EMS for Person Effect:

The model is $X_{pi} = \mu + \mu_p + \mu_i + \mu_{pi} + e_{pi}$.

The person sum of squares is:

$$SS_p = n_i \sum_{p=1}^{n_p} (\bar{X}_p - \bar{X})^2$$

where $\bar{X}_p = \frac{1}{n_i} \sum_i X_{pi}$ and $\bar{X}$ is the grand mean.

The expected value of SS_p requires computing $E(\bar{X}_p - \bar{X})^2$.

First, express $\bar{X}_p$ in terms of the model:

$$\bar{X}_p = \mu + \mu_p + \bar{\mu}_i + \bar{\mu}_{pi} + \bar{e}_p$$

where bars indicate averaging over items.

For random effects, $E(\mu_p^2) = \sigma_p^2$, $E(\mu_i^2) = \sigma_i^2$, $E(\mu_{pi}^2) = \sigma_{pi}^2$, and $E(e_{pi}^2) = \sigma_e^2$.

After algebraic manipulation:

$$E(MS_p) = E\left(\frac{SS_p}{n_p - 1}\right) = \sigma_e^2 + n_i \sigma_{pi}^2 + n_i \sigma_p^2$$

The key insight is that the person mean square contains variance from three sources: residual error, the person $\times$ item interaction (multiplied by n_i because each person mean averages over n_i items), and person variance (also multiplied by n_i).

Why the EMS has those three terms — the intuition. When you compute the variance of a person's mean across n_i items, the random part of the score gets averaged out. Specifically: residual error (σ_e^2) shrinks by a factor of n_i because each person sees n_i items and the errors partially cancel. The person $\times$ item interaction also shrinks by a factor of n_i , because the interaction is a random component that varies from item to item and therefore averages out across items. The *person* effect, however, is *constant* across items — so it does not shrink at all in the mean; it is multiplied by n_i in the mean square (a mean square is the sum of squares divided by df, and the sum of squares of a constant effect across n_i items scales as n_i). So $E(MS_p)$ is $\sigma_e^2 + n_i \sigma_{pi}^2 + n_i \sigma_p^2$: each piece of the equation is a different combination of “shrinks by averaging” and “stays constant across items.”

Solving EMS Equations

Given the EMS, we solve for variance components algebraically:

1. Start from the bottom of the ANOVA table (highest-order interaction)
2. Subtract appropriate mean squares to isolate components

For the $p \times i$ design:

$$\hat{\sigma}_{pi}^2 = \frac{MS_{pi} - MS_e}{1}$$

In practice, MS_{pi} is the lowest-order term, so:

$$\hat{\sigma}_{pi}^2 = MS_{pi} - MS_{residual}$$

But in balanced designs, we estimate:

$$\hat{\sigma}_{pi}^2 = MS_{pi} - MS_{residual}$$

For person variance:

$$\hat{\sigma}_p^2 = \frac{MS_p - MS_{pi}}{n_i}$$

For item variance:

$$\hat{\sigma}_i^2 = \frac{MS_i - MS_{pi}}{n_p}$$

EMS for Three-Facet Designs

Note. The EMS table below is for a **fully crossed** Person $\times$ Item $\times$ Rater design in which every rater rates every person on every item. The **brennan** dataset analyzed later in this vignette is *not* fully crossed (Rater is nested within Task), so its EMS table is different; see the next subsection.

For a fully crossed person $\times$ item $\times$ rater design ($p \times i \times r$), the EMS becomes more complex:

Source	EMS
Person (p)	$\sigma_e^2 + n_r\sigma_{pir}^2 + n_i\sigma_{pr}^2 + n_r\sigma_{pi}^2 + n_i n_r\sigma_p^2$
Item (i)	$\sigma_e^2 + n_r\sigma_{pir}^2 + n_p\sigma_{ir}^2 + n_r\sigma_{pi}^2 + n_p n_r\sigma_i^2$
Rater (r)	$\sigma_e^2 + n_i\sigma_{pir}^2 + n_p\sigma_{ir}^2 + n_i\sigma_{pr}^2 + n_p n_i\sigma_r^2$
$p \times i$	$\sigma_e^2 + n_r\sigma_{pir}^2 + n_r\sigma_{pi}^2$
$p \times r$	$\sigma_e^2 + n_i\sigma_{pir}^2 + n_i\sigma_{pr}^2$
$i \times r$	$\sigma_e^2 + n_p\sigma_{pir}^2 + n_p\sigma_{ir}^2$
$p \times i \times r$	$\sigma_e^2 + \sigma_{pir}^2$

The solving procedure follows the same pattern, starting from the three-way interaction.

EMS for the Nested brennan Design

The **brennan** data is a Person $\times$ Task $\times$ (Rater:Task) design — Person is crossed with Task, but Rater is nested within Task, with n_r raters per task. The model is

$$X_{ptr} = \mu + \tau_p + \alpha_t + \rho_{r(t)} + (\tau\alpha)_{pt} + e_{ptr}$$

where e_{ptr} absorbs the Person $\times$ Rater:Task interaction (each cell has a single observation, so that effect is confounded with the residual). The EMS table for a balanced nested design with n_p persons, n_t tasks, and n_r raters per task is:

Source	df	EMS
Person (p)	$n_p - 1$	$\sigma_e^2 + n_r\sigma_{pt}^2 + n_t n_r\sigma_p^2$
Task (t)	$n_t - 1$	$\sigma_e^2 + n_r\sigma_{pt}^2 + n_p\sigma_{r:t}^2 + n_p n_r\sigma_t^2$
Rater within Task ($r : t$)	$n_t(n_r - 1)$	$\sigma_e^2 + \sigma_{pt}^2 + n_p\sigma_{r:t}^2$
Person $\times$ Task (pt)	$(n_p - 1)(n_t - 1)$	$\sigma_e^2 + n_r\sigma_{pt}^2$
Residual ($e \equiv p \times r : t$)	$n_t(n_p - 1)(n_r - 1)$	σ_e^2

Solving the EMS equations (Henderson's Method I) for the nested design, in the same bottom-up order:

$$\begin{aligned} \hat{\sigma}_e^2 &= MS_e, & \hat{\sigma}_{pt}^2 &= \frac{MS_{pt} - MS_e}{n_r}, & \hat{\sigma}_{r:t}^2 &= \frac{MS_{r:t} - MS_e - \hat{\sigma}_{pt}^2}{n_p} \\ \hat{\sigma}_p^2 &= \frac{MS_p - n_r\hat{\sigma}_{pt}^2 - MS_e}{n_t n_r}, & \hat{\sigma}_t^2 &= \frac{MS_t - n_p\hat{\sigma}_{r:t}^2 - n_r\hat{\sigma}_{pt}^2 - MS_e}{n_p n_r} \end{aligned}$$

EMS for Nested Designs

When facets are nested (e.g., items nested within subtests), the EMS structure changes. For items (i) nested within subtests (s):

Source	EMS
Subtest (s)	$\sigma_e^2 + n_i \sigma_{pi:s}^2 + n_p \sigma_{i:s}^2 + n_p n_i \sigma_s^2$
Items within subtest ($i : s$)	$\sigma_e^2 + n_p \sigma_{i:s}^2$
Person $\times$ subtest ($p \times s$)	$\sigma_e^2 + n_i \sigma_{pi:s}^2$
Person $\times$ item:subtest ($pi : s$)	$\sigma_e^2 + \sigma_{pi:s}^2$

Note that nested effects do not appear in EMS for crossed effects, simplifying the equations.

Method of Moments (MOM)

Historical Development of ANOVA-Based Estimation

The method of moments approach to variance component estimation has deep roots in statistical methodology. **Henderson (1953)** developed three methods (Methods I, II, III) for variance component estimation from ANOVA, providing the foundational approach for balanced and unbalanced designs. Henderson's Method I is the standard ANOVA approach used in balanced G-theory designs.

Searle, Casella, & McCulloch (1992) provided the definitive modern treatment in *Variance Components*, including comprehensive coverage of ANOVA methods, the problem of negative estimates, and the relationship to likelihood-based approaches.

Hartley & Rao (1967) established the connection between ANOVA and maximum likelihood methods, bridging classical and modern approaches to variance component estimation.

Satterthwaite (1941, 1946) developed the approximation for confidence intervals on variance components, treating linear combinations of mean squares as scaled chi-square variates. This approximation remains the standard for MOM confidence intervals and is implemented in the **facet** package.

Brennan (2001) synthesized these methods specifically for G-theory, providing the rules for constructing Expected Mean Squares (EMS) that are implemented in the MOM backend.

Mathematical Foundation

The Method of Moments, also known as ANOVA-based estimation, derives variance components from Expected Mean Squares (EMS). For a balanced two-facet crossed design ($p \times i$), the ANOVA table yields:

Source	MS	EMS
Person (p)	MS_p	$\sigma_e^2 + n_i \sigma_{pi}^2 + n_i \sigma_p^2$
Item (i)	MS_i	$\sigma_e^2 + n_p \sigma_{pi}^2 + n_p \sigma_i^2$
$p \times i$	MS_{pi}	$\sigma_e^2 + \sigma_{pi}^2$

Solving the system of EMS equations:

$$\hat{\sigma}_{pi}^2 = \frac{MS_{pi} - MS_e}{n}$$

$$\hat{\sigma}_p^2 = \frac{MS_p - MS_{pi}}{n_i}$$

$$\hat{\sigma}_i^2 = \frac{MS_i - MS_{pi}}{n_p}$$

For unbalanced designs, the EMS equations become more complex. The **facet** package detects crossing patterns in the data and computes appropriate effective sample sizes.

Henderson's Method I Worked Example

Henderson's Method I (Henderson 1953) is the standard ANOVA approach used for balanced G-theory designs. The procedure is:

1. Compute the ANOVA table
2. Derive EMS for each source
3. Solve the system of EMS equations algebraically

Example: $p \times i$ design with 10 persons and 5 items

If $MS_p = 15.2$, $MS_i = 8.7$, and $MS_{pi} = 3.4$:

$$\hat{\sigma}_p^2 = \frac{15.2 - 3.4}{5} = \frac{11.8}{5} = 2.36$$

$$\hat{\sigma}_i^2 = \frac{8.7 - 3.4}{10} = \frac{5.3}{10} = 0.53$$

$$\hat{\sigma}_{pi}^2 = 3.4$$

The interpretation: universe score variance (person) is 2.36, item difficulty variance is 0.53, and the person $\times$ item interaction is 3.4. Total variance = 6.29, so person variance accounts for 37.5% of total variance.

Handling Negative Estimates

MOM can produce negative variance estimates when sample mean squares do not follow their expected ordering. For example, if by sampling error $MS_{pi} > MS_p$, the person variance estimate would be negative.

Standard practice: Truncate negative estimates to zero. This is ad hoc but practical. Negative estimates indicate either:

1. The true variance component is near zero
2. Sampling error in mean squares
3. Model misspecification

Confidence Intervals: Satterthwaite Approximation

(Satterthwaite 1941) and (Satterthwaite 1946) developed an approximate distribution for linear combinations of mean squares. For a variance component $\hat{\sigma}^2 = (MS_a - MS_b)/n$:

The approximate degrees of freedom are:

$$df_{approx} = \frac{(MS_a - MS_b)^2}{\frac{MS_a^2}{df_a} + \frac{MS_b^2}{df_b}}$$

The standard error is:

$$SE(\hat{\sigma}^2) \approx \sqrt{\frac{2 \cdot MS^2}{df}}$$

Confidence intervals use the chi-square distribution:

$$CI = \frac{df \cdot \hat{\sigma}^2}{\chi_{1-\alpha/2, df}^2} \text{ to } \frac{df \cdot \hat{\sigma}^2}{\chi_{\alpha/2, df}^2}$$

This chi-square-based CI is asymmetric and respects the boundary constraint (variance ≥ 0). The simpler Wald-type approximation $\hat{\sigma}^2 \pm z_{\alpha/2} \cdot SE(\hat{\sigma}^2)$ is also used but may produce negative lower bounds that are truncated to zero. Both work reasonably for large degrees of freedom but may be inaccurate for small samples.

A worked Satterthwaite example. Suppose you have a $p \times i$ design with $n_p = 10$ persons and $n_i = 5$ items, and you have $MS_p = 15.2$, $MS_{pi} = 3.4$, $df_p = 9$, $df_{pi} = 36$. The person variance estimate is $\hat{\sigma}_p^2 = (15.2 - 3.4)/5 = 2.36$. The approximate degrees of freedom are:

$$df_{approx} = \frac{(15.2 - 3.4)^2}{(15.2^2/9) + (3.4^2/36)} = \frac{139.1}{25.7 + 0.32} \approx 5.4$$

The 95% CI uses $\chi_{0.975, 5.4}^2$ and $\chi_{0.025, 5.4}^2$:

$$CI = \left[\frac{5.4 \times 2.36}{12.83}, \frac{5.4 \times 2.36}{0.51} \right] \approx [0.99, 24.98]$$

Note how wide this CI is — about $[1.0, 25.0]$ — even though the point estimate is 2.36. The wide CI reflects two things: (1) the small sample size, and (2) the fact that we are estimating a difference of two mean squares, so the effective df is much smaller than the df of either MS alone. With $n_p = 50$ persons and $n_i = 20$ items, the CI would shrink to roughly $[1.6, 3.7]$. The practical lesson: **the Satterthwaite df is often the binding constraint on CI width, not the raw sample size.**

Strengths and Limitations

Strengths:

- Fast computation (closed-form solution)
- No distributional assumptions beyond finite variance
- Directly interpretable ANOVA decomposition
- Asymptotically unbiased for balanced designs

Limitations:

- Requires balanced designs for unbiased estimates
- Can produce negative variance estimates (truncated to zero)
- Confidence intervals are approximate
- Limited support for complex nested designs
- No mechanism for incorporating prior knowledge

Implementation in facet

The `backend = "mom"` option uses R's `aov()` function with `Error()` strata:

```
# Method of moments G-study (canonical brennan formula; see
# "Specifying All Possible Variance Components")
g_mom <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                (1|Person:Task),
                data = brennan, backend = "mom")
```

The implementation:

1. Parses the formula to identify facets
2. Constructs an `aov()` formula with `Error()` terms
3. Extracts mean squares from each stratum
4. Detects crossing/nesting patterns from data
5. Solves EMS equations for variance components
6. Computes Satterthwaite approximate CIs

Restricted Maximum Likelihood (REML)

Historical Development of REML

Patterson & Thompson (1971) introduced Restricted Maximum Likelihood, recognizing that ML estimators of variance components are biased downward because they do not account for degrees of freedom lost in estimating fixed effects. Their approach partitions the likelihood into independent components: one containing information about variance components and another about fixed effects.

Harville (1974, 1977) established the theoretical foundation showing that REML estimators reduce bias compared to ML, with the bias reduction particularly important for small samples. Harville (1977) demonstrated that ML estimators of variance components are negatively biased, while REML estimators reduce or eliminate this bias.

Searle, Casella, & McCulloch (1992) provided comprehensive treatment of REML estimation, including computational algorithms and the relationship to ANOVA methods (in balanced designs, REML estimates equal ANOVA estimates).

Bates et al. (2015) developed the `lme4` package implementing efficient REML estimation using sparse matrix methods and Cholesky decomposition, enabling analysis of complex crossed designs that were previously computationally infeasible.

The `lme4` backend in **facet** leverages these advances, providing REML estimation with profile likelihood confidence intervals for variance components.

Mathematical Foundation

Restricted Maximum Likelihood (REML) estimates variance components by maximizing the likelihood of linear combinations of the data that eliminate fixed effects. For a linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \epsilon$$

Where:

- $\mathbf{y}$ is the response vector
- $\mathbf{X}$ is the fixed-effects design matrix
- β is the fixed-effects vector
- $\mathbf{Z}$ is the random-effects design matrix
- $\mathbf{u} \sim N(\mathbf{0}, \mathbf{G})$ are random effects
- $\epsilon \sim N(\mathbf{0}, \mathbf{R})$ is residual error

The variance-covariance matrix of $\mathbf{y}$ is:

$$\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R}$$

The REML log-likelihood is:

$$\ell_{REML}(\theta) = -\frac{1}{2} [\log |\mathbf{V}| + \mathbf{y}'\mathbf{P}\mathbf{y} + \log |\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}|]$$

Where $\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}$ is the projection matrix that removes fixed effects.

REML is preferred over ML because it accounts for degrees of freedom lost in estimating fixed effects, producing less biased variance component estimates.

Confidence Intervals

Profile likelihood CIs are constructed by inverting the likelihood ratio test:

$$CI = \{\theta : -2[\ell(\theta) - \ell(\hat{\theta})] < \chi_{1,1-\alpha}^2\}$$

Parametric bootstrap CIs are computed by:

1. Simulating new datasets from the fitted model
2. Re-fitting the model to each simulated dataset
3. Computing variance component estimates
4. Taking percentiles of the bootstrap distribution

Profile likelihood CIs are faster but may be unreliable for boundary parameters (variances near zero). Bootstrap CIs are more accurate but computationally intensive.

Strengths and Limitations

Strengths:

- Handles unbalanced designs gracefully
- Produces non-negative variance estimates
- Flexible modeling of complex designs
- Profile and bootstrap CI options
- Well-established asymptotic theory

Limitations:

- Slower than MOM for large datasets
- Convergence issues possible for complex models
- Cannot handle multivariate models (use brms instead)
- Variance estimates near zero can be problematic
- No mechanism for prior information

Implementation in facet

The default `backend = "lme4"` uses `lme4::lmer()` for estimation:

```
# REML G-study (default; canonical brennan formula; see
# "Specifying All Possible Variance Components")
g_lme4 <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                (1|Person:Task),
                data = brennan)

# With profile likelihood CIs
g_profile <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                   (1|Person:Task),
                   data = brennan, ci_method = "profile")

# With bootstrap CIs
g_boot <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                 (1|Person:Task),
                 data = brennan, ci_method = "boot", nsim = 100)
```

The implementation:

1. Converts formula to `lme4` syntax
2. Fits model using `lmer()`
3. Extracts variance components via `VarCorr()`
4. Optionally computes CIs via `confint.merMod()`

Bayesian Estimation with NUTS/HMC

Bayesian Advances in Generalizability Theory

A growing body of research has advocated for Bayesian estimation of G-theory models, providing full posterior distributions for variance components and G-coefficients rather than point estimates. **Putka &**

Hoffman (2013) pioneered the application of Bayesian G-theory to assessment center data, demonstrating advantages over ANOVA-based approaches. **LoPilato, Carter, & Wang (2015)** provided foundational simulation evidence for prior selection and posterior summarization in management research.

Jackson and colleagues applied properly specified Bayesian G-theory models to assessment center and performance rating data, addressing a specific problem in the applied literature: prior AC research had often used *mis-specified* G-theory models that omitted interaction terms, leading to confounded variance component estimates. In a properly specified G-theory model, all relevant interaction terms are included by design (Cronbach et al. (1972); Brennan (2001)). Jackson et al.’s contributions were twofold:

1. **Substantive findings:** Using Bayesian G-theory with fully crossed Person $\times$ Exercise $\times$ Rater designs, they showed that dimension-related variance was negligibly small in AC ratings (Jackson et al. (2016)), a finding replicated with multisource performance ratings (Jackson et al. (2020)). Subsequent work showed that reliability in ACs depends on general and exercise effects, but not on dimensions ((**jackson2022b?**))), and that uncertainty about whether rater variance constitutes signal or error fundamentally impacts reliability estimates for supervisor ratings (Jackson et al. (2022)).
2. **Interpretive reframing:** Jackson et al. (2023) demonstrated that G-theory random effects models and CFA models yield equivalent variance components, generalizability coefficients, and latent scores, arguing that G-theory can address both reliability *and* structural validity questions—not just reliability as traditionally assumed.

Jiang & Skorupski (2018) developed Bayesian estimation methods for multivariate G-theory, handling variance-covariance matrices and providing uncertainty quantification for multivariate designs through BUGS/JAGS implementations.

Full Crossed Designs with Bayesian Estimation

In crossed designs where all facet combinations are observed, a properly specified G-theory model includes all *estimable* interaction terms. For example, a fully crossed Person $\times$ Task $\times$ Rater design with multiple observations per Person $\times$ Task $\times$ Rater cell would include all two-way and three-way interactions:

```
# Fully crossed design with REPLICATION per cell (e.g. each rater rates
# every person on every task on multiple occasions). The three-way
# Person:Task:Rater interaction IS estimable here because each cell has
# more than one observation.
g_crossed <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task) + (1 | Person:Rater) + (1 | Task:Rater) +
    (1 | Person:Task:Rater),
  data = crossed_data,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)
```

The **brennan** dataset is **not** such a design: it is Rater nested within Task with one observation per Person $\times$ Rater:Task cell, so the three-way interaction is confounded with the residual and is absorbed into the error term. For **brennan**, the canonical Bayesian specification (per “Specifying All Possible Variance Components”) is:

```
g_nested <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task),
  data = brennan,
```

```

backend = "brms",
iter = 2000, cores = 4, refresh = 1000
)

```

Omitting interaction terms from a crossed design constitutes model mis-specification—the omitted interaction variance is absorbed by the included terms, producing confounded estimates. This is not a limitation of G-theory itself but a consequence of under-specifying the model. Bayesian estimation via `backend = "brms"` provides full posterior distributions for each variance component, including interactions, enabling credible intervals for G-coefficients and D-study projections.

The `facet` package’s `backend = "brms"` option implements these methodological advances through the Stan probabilistic programming language (Bürkner, 2017), providing researchers with tools for Bayesian G-theory that properly handle variance component estimation in complex designs.

Mathematical Foundation

Bayesian estimation treats variance components as random variables with prior distributions. The posterior distribution is:

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta) \cdot p(\theta)$$

Where $p(\theta)$ is the prior and $p(\mathbf{y}|\theta)$ is the likelihood.

Prior distributions for variance components must be positive-valued. Common choices:

- **Half-Cauchy:** $p(\sigma) \propto \frac{1}{1+\sigma^2/s^2}$ (recommended for variance parameters)
- **Exponential:** $p(\sigma) = \lambda e^{-\lambda\sigma}$ (simple, one-parameter)
- **Inverse-Gamma:** $p(\sigma^2) \propto (\sigma^2)^{-(\alpha+1)} e^{-\beta/\sigma^2}$ (traditional but problematic)

The half-Cauchy prior is recommended because it has heavy tails (allowing large variance values) while concentrating probability mass near zero for small effects.

Hamiltonian Monte Carlo (HMC) is a gradient-based MCMC algorithm that exploits the geometry of the posterior distribution. It treats the negative log-posterior as a potential energy surface and simulates Hamiltonian dynamics:

$$H(\theta, \mathbf{p}) = U(\theta) + K(\mathbf{p})$$

Where $U(\theta) = -\log p(\theta|\mathbf{y})$ is potential energy and $K(\mathbf{p}) = \frac{1}{2}\mathbf{p}'\mathbf{M}^{-1}\mathbf{p}$ is kinetic energy.

The **No-U-Turn Sampler (NUTS)** automatically tunes HMC parameters:

- Step size ϵ (adapted during warmup)
- Number of leapfrog steps L (determined dynamically)
- Mass matrix $\mathbf{M}$ (adapted to posterior geometry)

The HMC intuition. Imagine the posterior distribution as a landscape of varying elevation, where height = negative log-posterior density. Standard MCMC (e.g., Metropolis-Hastings) takes small random steps that can get stuck in narrow valleys or wander aimlessly across high-density plateaus. HMC treats the chain as a *frictionless particle rolling on this landscape* with a randomly assigned momentum. The particle is allowed to roll for some number of leapfrog steps, then its position is recorded as a new sample. The leapfrog integrator approximately conserves Hamiltonian dynamics, so the sampler moves in long, informed

trajectories that efficiently explore the posterior’s geometry. The NUTS extension automates the choice of trajectory length: it stops when the particle starts to “double back” on itself (a “U-turn”), so the user does not have to specify the number of leapfrog steps. The practical payoff for G-theory: HMC handles the *ratios* of variance components (G, Phi) gracefully, because the full posterior is sampled — there is no need for the delta-method or bootstrap approximations that frequentist CIs require for derived quantities.

Convergence Diagnostics

R-hat ($\hat{R}$) compares within-chain and between-chain variance:

$$\hat{R} = \sqrt{\frac{\widehat{var}(\psi)}{W}}$$

Where $\widehat{var}(\psi)$ combines within-chain (W) and between-chain (B) variance. Values near 1.0 indicate convergence; values > 1.01 suggest non-convergence.

Effective Sample Size (ESS) estimates the number of independent samples:

$$ESS = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

Where ρ_k is the autocorrelation at lag k . Higher ESS indicates better mixing. Target: $ESS > 400$ for reliable inference.

Posterior Inference

Bayesian inference provides:

- **Posterior mean:** $E[\sigma^2|\mathbf{y}]$ (point estimate)
- **Posterior SD:** $SD[\sigma^2|\mathbf{y}]$ (uncertainty)
- **Credible intervals:** $[\sigma_{0.025}^2, \sigma_{0.975}^2]$ (95% CI)
- **Full posterior distribution:** Samples from $p(\sigma^2|\mathbf{y})$

For D-study coefficients, the posterior distribution propagates variance component uncertainty:

$$p(g|\mathbf{y}) = \int p(g|\sigma^2) \cdot p(\sigma^2|\mathbf{y}) d\sigma^2$$

This naturally incorporates uncertainty into coefficient estimates.

Strengths and Limitations

Strengths:

- Full uncertainty quantification via posterior distributions
- Custom prior distributions for domain knowledge
- Handles multivariate models with correlated random effects
- No asymptotic approximations needed
- Coherent inference for derived quantities (G, Phi coefficients)
- Model comparison via LOO-CV, Bayes factors

Limitations:

- Computationally intensive (minutes to hours)
- Requires convergence diagnostics
- Prior sensitivity analysis recommended
- Technical expertise needed for prior specification
- Model compilation overhead for complex models

Implementation in facet

The `backend = "brms"` option uses `brms::brm()` which compiles models to Stan:

```
# Bayesian G-study with default priors (canonical brennan formula; see
# "Specifying All Possible Variance Components")
g_brms <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                 (1|Person:Task),
                 data = brennan, backend = "brms",
                 iter = 2000, cores = 4, refresh = 1000)

# With custom priors
library(brms)
my_prior <- c(
  set_prior("exponential(1)", class = "sd", group = "Person"),
  set_prior("exponential(0.5)", class = "sd", group = "Task"),
  set_prior("exponential(1)", class = "sd", group = "Rater"),
  set_prior("exponential(1)", class = "sd", group = "Person:Task"),
  set_prior("exponential(1)", class = "sigma")
)
g_custom <- gstudy(Score ~ (1|Person) + (1|Task) + (1|Rater) +
                  (1|Person:Task),
                  data = brennan, backend = "brms", prior = my_prior,
                  iter = 2000, cores = 4, refresh = 1000)

# Multivariate Bayesian G-study (requires a fully crossed design;
# see the `crossed_wide` data defined in the Multivariate G-Study section)
g_mv <- gstudy(mvbind(Task1, Task2, Task3) ~ (1|Person) + (1|Rater),
               data = crossed_wide, backend = "brms",
               iter = 2000, cores = 4, refresh = 1000)
```

The implementation:

1. Converts formula to `brms` syntax
2. Compiles model to Stan
3. Runs NUTS sampling (default: 4 chains, 2000 iterations)
4. Extracts posterior draws for variance components
5. Computes posterior summaries and diagnostics

When to Use Bayesian Methods

Bayesian estimation is particularly valuable when:

Situation	Why Bayesian Helps
Multivariate designs	Naturally handles variance-covariance matrices
Small samples	Prior information stabilizes estimates
Complex designs	Avoids asymptotic approximations
Derived quantities	Propagates uncertainty through transformations
Custom priors	Incorporates domain expertise
Decision analysis	Direct probability statements support decisions

Prerequisites

The Bayesian backend requires additional packages:

```
# Core package
library(facet)

# Bayesian estimation
library(brms)

# Posterior manipulation
library(posterior)

# Visualization
library(ggplot2)
library(bayesplot)
```

Default Priors in brms

For variance components (standard deviations), brms uses weakly informative priors by default:

$$\sigma \sim \text{Student-}t_3(0, \sigma_{\text{data}})$$

This prior:

- Is bounded below at zero (appropriate for standard deviations)
- Has heavy tails (allows large values if supported by data)
- Is weakly informative (centered at the scale of the data)

Recommended Priors for Variance Components

Half-Cauchy Prior (Gelman 2006):

$$\sigma \sim \text{Half-Cauchy}(0, s)$$

With density:

$$p(\sigma) = \frac{2}{\pi s (1 + (\sigma/s)^2)}, \quad \sigma > 0$$

Properties: - Heavy tails allow large variance values - Concentrates mass near zero for small effects - Scale parameter s controls spread

Exponential Prior:

For simplicity, the exponential prior provides a one-parameter alternative:

$$\sigma \sim \text{Exponential}(\lambda)$$

With rate parameter λ :

$$p(\sigma) = \lambda e^{-\lambda\sigma}, \quad \sigma > 0$$

The expected value is $E[\sigma] = 1/\lambda$. Setting $\lambda = 1$ gives a prior centered at $\sigma = 1$.

Why half-Cauchy is preferred over inverse-gamma. The traditional prior on variance parameters (i.e., on σ^2 , not σ) is the inverse-gamma family. It has a serious problem: the inverse-gamma has a pole at zero that pulls the *posterior* away from zero, even when the data want it there. This is a well-documented pathology in mixed models — when the true variance component is small, an inverse-gamma prior can keep the posterior artificially away from zero, leading to overestimation. The half-Cauchy, by contrast, has a *cusp* at zero: it concentrates mass near zero (good for small effects) but has heavy tails that allow large values if supported by the data. See Gelman (2006) for the full argument. As a practical rule: **always prefer the half-Cauchy or Student-t family for variance parameters; avoid inverse-gamma unless you have strong prior information that the variance is bounded away from zero.**

Regularizing Priors (code examples):

```
# Weakly informative (recommended for most applications)
prior_weak <- c(
  set_prior("cauchy(0, 2)", class = "sd"),
  set_prior("cauchy(0, 2)", class = "sigma")
)

# Slightly informative (when some variance components expected small)
prior_reg <- c(
  set_prior("exponential(1)", class = "sd"),
  set_prior("exponential(1)", class = "sigma")
)

# Strongly regularizing (for very small samples or complex designs)
prior_strong <- c(
  set_prior("exponential(2)", class = "sd"),
  set_prior("exponential(2)", class = "sigma")
)
```

Prior for Fixed Effects

For the grand mean (intercept), default priors are typically uninformative:

$$\mu \sim \mathcal{N}(0, 10)$$

Or more informative if the scale of measurement is known:

```
# If scores are expected around 5 on a 1-10 scale
prior_mean <- set_prior("normal(5, 2)", class = "Intercept")
```

Why HMC matters specifically for G-theory. G-theory produces *derived quantities* — G and Phi coefficients — that are nonlinear functions (ratios) of the variance components. In frequentist estimation, the only way to get a CI for a ratio is the delta method (a linear approximation that can be inaccurate for small samples) or the bootstrap (computationally expensive and noisy with few resamples). HMC's full posterior samples let you compute the ratio for *every* posterior draw, giving exact credible intervals for G and Phi with no asymptotic approximations. The credible intervals you see from `dstudy(..., ci = c("g", "phi"))` are computed this way: each draw of the variance components produces a draw of G, and the 2.5th/97.5th percentiles of those G draws are the CI limits. This is one of the main reasons to use Bayesian estimation when you need defensible uncertainty quantification for reliability coefficients.

```
# If scores are expected around 5 on a 1-10 scale
prior_mean <- set_prior("normal(5, 2)", class = "Intercept")
```

Comparison of Estimation Methods

Criterion	MOM	REML (lme4)	Bayesian (brms)
Speed	Fast (< 1 sec)	Moderate (1-10 sec)	Slow (minutes)
Balanced data	Required	Not required	Not required
Unbalanced data	Approximate	Exact	Exact
Uncertainty	Asymptotic CI	Profile/Bootstrap CI	Full posterior
Negative variance	Possible (truncated)	Constrained to 0+	Constrained to 0+
Custom priors	N/A	N/A	Yes
Multivariate	Limited	No	Full support
Diagnostics	None	Convergence warnings	R-hat, ESS
Complex designs	Limited	Good	Excellent

Recommendations:

- Use **MOM** for quick exploratory analysis of balanced designs
- Use **REML** (default) for routine univariate analysis with unbalanced data
- Use **Bayesian** for multivariate models, custom priors, or full uncertainty quantification

Using the facet Package

Contemporary Applications of Generalizability Theory

Since its development, G-theory has expanded dramatically across diverse fields:

Educational Measurement and Large-Scale Assessment

G-theory is foundational for large-scale educational assessments including NAEP, PISA, and TIMSS. Researchers use G-theory to examine rater effects in essay scoring, evaluate test form comparability, and optimize assessment designs (Briesch, Chafouleas, & Machek, 2016).

Cross-cultural applications examine measurement equivalence across countries. **Soysal (2023)** demonstrated treating culture as a measurement facet in G-theory analyses of PISA mathematics attitude scales, while **Solano-Flores & Milbourn (2016)** argued for G-theory to examine cultural validity in international assessments.

Performance Assessment

Writing assessment, teacher evaluation, and portfolio assessment routinely employ G-theory to understand rater effects and optimize scoring designs. **Atılgan (2019)** demonstrated optimal rater allocation strategies for essay scoring, finding 3-4 raters optimal for acceptable reliability.

Casale, Strauß, Hennemann, & König (2016) applied G-theory to video-based teacher performance assessment, identifying key sources of variance in teacher evaluation ratings.

Medical and Healthcare Education

The most dramatic growth area has been competency-based medical education, particularly Objective Structured Clinical Examinations (OSCEs). **Andersen, Nayahangan, Park, & Konge (2021)** conducted a systematic review and meta-analysis of 39 studies using G-theory for technical skills assessment, identifying optimal designs for procedural competence evaluation.

Trejo-Mejía, Sánchez-Mendiola, et al. (2016) analyzed OSCE reliability in Latin American medical education, demonstrating G-theory's value for clinical skills assessment across cultural contexts.

Organizational Psychology

Assessment center ratings and 360-degree feedback systems use G-theory to understand dimension variance and exercise effects. **Merkulova, Melchers, & Kleinmann (2016)** used G-theory to examine variance components in assessment center ratings, contributing to the debate about construct validity in personnel selection.

Cross-Cultural and Multilingual Assessment

G-theory provides the framework for examining measurement equivalence in multilingual assessments. **Oliveri, Ercikan, & Simon (2015)** developed frameworks for multilingual test development using G-theory, while **Solano-Flores, Trumbull, & Nelson-Barber (2011)** demonstrated G-theory applications for English Language Learner assessment.

Installation

```
# Install from GitHub
# install.packages("devtools")
devtools::install_github("michaelides/facet")

# Load required packages
library(facet)
library(dplyr)
library(tidyr)
library(broom)
library(ggplot2)
```

Note: The `tidy()` and `glance()` methods require the `broom` package to be loaded.

Built-in Datasets

The facet package includes two classic G-theory datasets:

```
library(facet)
library(broom)
library(dplyr)

# Brennan (2001) dataset: Person crossed with Task and Rater
data(brennan)
head(brennan)
#>   Task Person Rater Score y
#> 1     1      1      1      5 5
#> 2     1      2      1      9 9
#> 3     1      3      1      3 3
#> 4     1      4      1      7 7
#> 5     1      5      1      9 9
#> 6     1      6      1      3 3

# Rajaratnam, Cronbach, & Gleser (1965): Person crossed with Subtest,
# Items nested within Subtest
data(rajaratnam)
head(rajaratnam)
#>   Person Subtest Item Score
#> 1      1      1      1      4
#> 2      2      1      1      2
#> 3      3      1      1      2
#> 4      4      1      1      1
#> 5      5      1      1      3
#> 6      6      1      1      1
```

The **brennan** dataset contains scores for 10 persons across 3 tasks, with a panel of 4 raters nested within each task (12 raters in total). The design is $p \times t \times (r : t)$: Person is crossed with Task, and Rater is nested within Task (raters 1–4 only see Task 1, raters 5–8 only see Task 2, and raters 9–12 only see Task 3). This is ideal for demonstrating nested designs.

Design note (brennan). The canonical G-study formula for the brennan data (per the rule in “Specifying All Possible Variance Components”) is

$$\text{Score} \sim (1 \mid \text{Person}) + (1 \mid \text{Task}) + (1 \mid \text{Rater}) + (1 \mid \text{Person:Task})$$

The four estimable random-effect variances are σ_{Person}^2 (universe-score variance), σ_{Task}^2 (task-difficulty variance), σ_{Rater}^2 (within-task rater-lenieny variance, equivalent to $\sigma_{\text{Rater:Task}}^2$ because the 12 **Rater** labels are unique within each **Task**), and $\sigma_{\text{Person:Task}}^2$ (person-by-task interaction). The residual absorbs the person-by-rater:task interaction (each **Person** $\times$ **Rater:Task** cell has a single observation, so that effect is confounded with the residual). Formally $(1 \mid \text{Rater})$ and $(1 \mid \text{Rater:Task})$ produce the same fitted model here—we use $(1 \mid \text{Rater})$ so the formula reads as “all main effects + the only estimable interaction” and so that downstream **dstudy()** calls can refer to the facet by the simple name **Rater**.

The **rajaratnam** dataset contains scores for 8 persons across 3 subtests, with items nested within subtest. The dataset ships with two item-related columns: the original **Item** factor (4 levels, with labels reused across subtests) and a disambiguated **ItemId** factor (8 levels, formed as `paste(Subtest, Item, sep =`

"."). The `ItemId` column is what the canonical G-study formula uses, because it makes the within-subtest nesting of items visible in the formula. Because `ItemId` is unique within `Subtest`, `(1 | ItemId)` and `(1 | Item:Subtest)` are exactly equivalent random-effect specifications—we use `(1 | ItemId)` for the same reason we use `(1 | Rater)` for `brennan`: it makes the “all main effects + the only estimable interaction” structure explicit and lets `dstudy()` refer to the facet as `ItemId = n_items`. This is ideal for demonstrating nested designs.

Basic G-Study

A G-study estimates variance components for each facet in the measurement design. The formula syntax follows `lme4` conventions, specifying random effects with `(1 | facet)`. We use the canonical `brennan` formula (per “Specifying All Possible Variance Components”):

```
# Canonical G-study for the brennan data: all main effects plus
# the only estimable interaction (Person:Task). The Person x Rater:Task
# interaction is confounded with the residual and absorbed into the
# error term.
g <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
            (1 | Person:Task),
            data = brennan)

# Print basic results
print(g)
#> Generalizability Study (G Study)
#> =====
#>
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#> Number of observations: 120
#> Multivariate: No
#>
#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect          type      n
#>   <chr>          <chr>   <dbl>
#> 1 Person        main      10
#> 2 Task          main       3
#> 3 Rater         main      12
#> 4 Person:Task   interaction  30
#> 5 Person:Rater (res) residual  120
#> 6 Rater per Task nested      4
#>
#> Nested details:
#>   Rater per Task: 3 Tasks, 4.0 Raters per Task
#>
#> Variance Components:
#> # A tibble: 5 x 3
#>   component estimate    pct
#>   <chr>          <dbl>   <dbl>
#> 1 Person        0.473  10.7856
```

```
#> 2 Task          0.3253  7.418
#> 3 Rater          0.6475 14.7637
#> 4 Person:Task    0.5596 12.7586
#> 5 Residual       2.3803 54.2741
```

The output shows the estimated variance components:

- **Person:** Variance attributable to differences between persons (universe score variance)
- **Task:** Variance attributable to differences in task difficulty
- **Rater:** Variance attributable to differences between raters (within-task rater leniency)
- **Person:Task:** Person-by-task interaction
- **Residual:** Remaining variance (person $\times$ rater:task interaction + error)

Viewing Results

Use `summary()` for a detailed view with options to display on variance or SD scale:

```
# Summary with variance estimates (default)
summary(g)
#> === G-Study Summary ===
#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:
#> Object of measurement: Person
#> Facets: Person, Task
#>
#> Sample Sizes:
#> # A tibble: 3 x 3
#>   effect      type      n
#>   <chr>      <chr>   <int>
#> 1 Person    main      10
#> 2 Task      main       3
#> 3 Person:Task (res) residual 30
#>
#> Variance Components:
#> # A tibble: 3 x 3
#>   component estimate  pct
#>   <chr>      <dbl> <dbl>
#> 1 Person      0.584  13.3
#> 2 Task        0.52  11.8
#> 3 Residual    3.29  74.9
#>
#> Total variance: 4.397

# Summary with standard deviation estimates
summary(g, scale = "sd")
#> === G-Study Summary ===
```



```

#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:
#> Object of measurement: Person
#> Facets: Person, Task
#>
#> Sample Sizes:
#> # A tibble: 3 x 3
#>   effect          type      n
#>   <chr>          <chr>  <int>
#> 1 Person        main      10
#> 2 Task          main       3
#> 3 Person:Task (res) residual  30
#>
#> Variance Components:
#> # A tibble: 3 x 3
#>   component estimate  pct
#>   <chr>          <dbl> <dbl>
#> 1 Person        0.764  13.3
#> 2 Task          0.721  11.8
#> 3 Residual      1.82   74.9
#>
#> Total variance: 4.397

```

Extract variance components as a tibble using `tidy()`:

```

tidy(g)
#> # A tibble: 3 x 5
#>   component dim type      var  pct
#>   <chr>    <chr> <chr>   <dbl> <dbl>
#> 1 Person  Score main    0.584  13.3
#> 2 Task    Score main    0.52   11.8
#> 3 Residual Score residual 3.29   74.9

```

Use `glance()` for a one-row summary of the G-study:

```

glance(g)
#> # A tibble: 1 x 5
#>   n_obs n_facets backend is_multivariate object
#>   <int>   <int> <chr>    <lgl>          <chr>
#> 1   120     2 lme4    FALSE          Person

```

Variance Component Percentages

The `pct` column shows what percentage of total variance each component accounts for. This helps identify which facets contribute most to measurement error:

```
# Components ordered by variance
tidy(g) %>%
  arrange(desc(var)) %>%
  mutate(pct_cumulative = cumsum(pct))
#> # A tibble: 3 x 6
#>   component dim   type      var  pct pct_cumulative
#>   <chr>      <chr> <chr>    <dbl> <dbl>         <dbl>
#> 1 Residual  Score residual 3.29   74.9          74.9
#> 2 Person   Score main    0.584  13.3          88.2
#> 3 Task     Score main    0.52   11.8          100
```

Interpreting variance component percentages. The `pct` column tells you where the measurement error is coming from. As a rough rule of thumb: if the Person component is less than 30% of total variance, the measurement is dominated by facets other than individual differences — meaning your instrument is “noisy” in the sense that it is hard to distinguish persons from one another. If a single facet (Task, Rater, etc.) accounts for more than 50% of variance, that facet is a clear target for design improvement: adding more levels of that facet will buy the largest gain in reliability. The cumulative percentage column is useful for identifying the top 2-3 sources of error — these are the facets worth investing in for your next measurement campaign.

G-Study with Confidence Intervals

For the `lme4` backend, you can request confidence intervals using the `ci_method` parameter.

Profile Likelihood Confidence Intervals

Profile likelihood CIs are more accurate than Wald-type intervals but slower to compute. Note: In the Brennan dataset, Raters are nested within Task (each Task has its own set of 4 raters), so we should not include crossed Rater effects with Task:

```
# G-study with profile likelihood confidence intervals
# Note: (1 | Rater) and (1 | Rater:Task) are equivalent in the brennan data
# because the 12 Rater labels are unique within each Task. We use the
# un-nested form so the formula reads as "all main effects + the only
# estimable interaction".
g_profile <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  ci_method = "profile"
)

summary(g_profile)
#> === G Study Summary ===
#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:
```

```

#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect      type      n
#>   <chr>      <chr>    <dbl>
#> 1 Person      main        10
#> 2 Task        main         3
#> 3 Rater       main        12
#> 4 Person:Task interaction   30
#> 5 Person:Rater (res) residual   120
#> 6 Rater per Task nested         4
#>
#> Nested details:
#>   Rater per Task: 3 Tasks, 4.0 Raters per Task
#>
#> Variance Components:
#> # A tibble: 5 x 5
#>   component estimate      pct `2.5% CI` `97.5% CI`
#>   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
#> 1 Person      0.473  10.7856     0      1.9449
#> 2 Task        0.3253   7.418     0      3.5622
#> 3 Rater       0.6475  14.7637   0.149    2.2824
#> 4 Person:Task  0.5596  12.7586     0      1.8827
#> 5 Residual    2.3803  54.2741   1.7758    3.2931
#>
#> Total variance: 4.3857

```

Bootstrap Confidence Intervals

Parametric bootstrap CIs are the most accurate but slowest:

```

# G-study with bootstrap confidence intervals (50 iterations for speed)
g_boot <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  ci_method = "boot",
  nsim = 50,
  boot.type = "perc"
)
#> =====

summary(g_boot)
#> === G-Study Summary ===
#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:

```

```

#> Object of measurement: Person
#> Facets: Person, Task
#>
#> Sample Sizes:
#> # A tibble: 3 x 3
#>   effect      type      n
#>   <chr>      <chr>   <int>
#> 1 Person      main      10
#> 2 Task        main       3
#> 3 Person:Task (res) residual  30
#>
#> Variance Components:
#> # A tibble: 3 x 5
#>   component estimate  pct `2.5% CI` `97.5% CI`
#>   <chr>      <dbl> <dbl>   <dbl>   <dbl>
#> 1 Person      0.584  13.3     0       1.35
#> 2 Task        0.52   11.8     0       1.63
#> 3 Residual     3.29  74.9    2.45    4.53
#>
#> Total variance: 4.397

```

For production use, increase `nsim` to 500-1000 for stable CI estimates.

Profile Likelihood vs Bootstrap: A Side-by-Side Comparison

The two CI methods answer the same question (where is the true variance component?) with different tradeoffs. The following chunk fits the same model with both methods and prints the CIs together:

```

# Fit the same model with both methods
g_profile <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  ci_method = "profile"
)

# Compare the CIs
profile_ci <- tidy(g_profile) %>%
  select(component, var, lower, upper) %>%
  mutate(method = "profile")

# Bootstrap CIs were computed above in g_boot
boot_ci <- tidy(g_boot) %>%
  select(component, var, lower, upper) %>%
  mutate(method = "boot (nsim=50)")

bind_rows(profile_ci, boot_ci) %>%
  select(method, component, var, lower, upper) %>%
  mutate(across(c(var, lower, upper), ~ sprintf("%.3f", .x)))
#> # A tibble: 8 x 5
#>   method      component  var  lower upper
#>   <chr>      <chr>      <chr> <chr> <chr>

```

```

#> 1 profile      Person      0.473 0.000 1.945
#> 2 profile      Task        0.325 0.000 3.562
#> 3 profile      Rater       0.647 0.149 2.282
#> 4 profile      Person:Task 0.560 0.000 1.883
#> 5 profile      Residual    2.380 1.776 3.293
#> 6 boot (nsim=50) Person     0.584 0.000 1.352
#> 7 boot (nsim=50) Task      0.520 0.000 1.629
#> 8 boot (nsim=50) Residual   3.293 2.446 4.532

```

Reading the comparison. In this small balanced design, the two methods usually agree closely on point estimates but can differ on CI width — particularly for components near zero, where the profile-likelihood CI may underestimate uncertainty. The bootstrap is more honest about boundary effects but is also noisier when `nsim` is small (here 50; for publication use 500+). A practical rule: use `ci_method = "profile"` for routine reporting in balanced or near-balanced designs, and `ci_method = "boot"` when components are near zero, when you want to defend a “variance is plausibly zero” claim, or when reviewers want conservative uncertainty quantification.

Method of Moments Backend

The Method of Moments backend uses ANOVA-based variance component estimation:

```

# Method of moments G-study with nested design (Rater nested in Task;
# (1 | Rater) and (1 | Rater:Task) are equivalent here)
g_mom <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "mom"
)

summary(g_mom)
#> === G Study Summary ===
#>
#> Design Information:
#> Backend: mom
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:
#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect      type      n
#>   <chr>      <chr>    <dbl>
#> 1 Person    main        10
#> 2 Task      main         3
#> 3 Rater     main        12
#> 4 Person:Task interaction  30
#> 5 Person:Rater (res) residual  120

```

```

#> 6 Rater per Task      nested      4
#>
#> Nested details:
#>   Rater per Task: 3 Tasks, 4.0 Raters per Task
#>
#> Variance Components:
#> # A tibble: 5 x 5
#>   component estimate      pct `2.5% CI` `97.5% CI`
#>   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
#> 1 Person          0.4731 10.4045      0      1.3048
#> 2 Task            0.487 10.7099      0      1.6703
#> 3 Rater           0.6475 14.2392      0      1.469
#> 4 Person:Task     0.5596 12.3049      0      1.3359
#> 5 Residual        2.3802 52.3415     1.6472     3.1133
#>
#> Total variance: 4.5474

```

MOM automatically provides asymptotic confidence intervals. This approach works best with balanced designs.

Nested Designs with MOM

For nested designs, the canonical formula uses the un-nested form when the nested-factor labels are already unique within the nesting factor. For `rajaratnam`, items are uniquely labelled within each subtest via the new `ItemId` column (formed as `paste(Subtest, Item, sep = ".")`), so `(1 | ItemId)` and `(1 | Item:Subtest)` are exactly equivalent (we use `(1 | ItemId)`):

```

# Items nested within subtests (rajaratnam dataset)
# (1 | ItemId) and (1 | Item:Subtest) are exactly equivalent here.
g_nested <- gstudy(
  Score ~ (1 | Person) + (1 | Subtest) + (1 | ItemId) +
    (1 | Person:Subtest),
  data = rajaratnam,
  backend = "mom"
)

summary(g_nested)
#> === G Study Summary ===
#>
#> Design Information:
#>   Backend: mom
#>   Formula: Score ~ (1 | Person) + (1 | Subtest) + (1 | ItemId) + (1 | Person:Subtest)
#>   Observations: 64
#>   Multivariate: No
#>
#> Facet Information:
#>   Object of measurement: Person
#>   Facets: Person, Subtest, ItemId
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect          type      n

```

```

#>   <chr>                <chr>          <dbl>
#> 1 Person              main            8
#> 2 Subtest             main            3
#> 3 ItemId              main            8
#> 4 Person:Subtest      interaction 24
#> 5 Person:ItemId (res) residual      64
#> 6 ItemId per Subtest  nested        2.67
#>
#> Nested details:
#>   ItemId per Subtest: 3 Subtests, 2.7 ItemIds per Subtest (harmonic: 2.4)
#>
#> Variance Components:
#> # A tibble: 5 x 5
#>   component      estimate      pct `2.5% CI` `97.5% CI`
#>   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
#> 1 Person        1.2634 20.4609      0      3.0486
#> 2 Subtest       2.8443 46.0642      0      8.7253
#> 3 ItemId        0.2946  4.7718      0      0.7929
#> 4 Person:Subtest 0.9295 15.0529      0      1.864
#> 5 Residual      0.8429 13.6502    0.448    1.2378
#>
#> Total variance: 6.1747

```

Bayesian G-Study

For Bayesian estimation, use the **brms** backend. This provides full posterior distributions for variance components.

Data Preparation

The **brennan** dataset provides a crossed design with persons, tasks, and raters:

```

library(facet)
data(brennan)

# Examine structure
head(brennan)
#   Person Task Rater Score
# 1      1    1     1     5
# 2      2    1     1     9
# 3      3    1     1     3
# 4      4    1     1     7
# 5      5    1     1     9
# 6      6    1     1     3

str(brennan)

```

Default Bayesian G-Study

The simplest Bayesian G-study uses default priors. The canonical **brennan** formula uses `(1 | Rater)` (equivalent to `(1 | Rater:Task)` because the 12 Rater labels are unique within each Task) and `(1 | Person:Task)`:

```
# Bayesian G-study with default priors
g_bayes <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

print(g_bayes)
summary(g_bayes)
```

Understanding the Output

The output includes:

Column	Description
component	Variance component name
estimate	Posterior mean
sd	Posterior standard deviation
lower	95% credible interval lower bound
upper	95% credible interval upper bound
Rhat	Convergence diagnostic (target: < 1.01)
Bulk_ESS	Effective sample size for posterior bulk
Tail_ESS	Effective sample size for posterior tails

Convergence diagnostics interpretation:

- **R-hat** ($\hat{R}$) compares within-chain and between-chain variance:

$$\hat{R} = \sqrt{\frac{\widehat{\text{var}}(\psi)}{W}}$$

Where $\widehat{\text{var}}(\psi) = \frac{n-1}{n}W + \frac{1}{n}B$ with W = within-chain variance and B = between-chain variance.

- Values < 1.01 indicate convergence
- Values > 1.05 suggest problems
- **Effective Sample Size (ESS)** estimates the number of independent samples:

$$\text{ESS} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

- Target: ESS > 400 for reliable inference
- Higher ESS indicates better mixing

Reading R-hat and ESS in practice. Compare the values *across* components, not just against fixed thresholds. A common pattern: R-hat and ESS are fine for the Person and main facet components, but one component (often a small or near-zero interaction) shows R-hat = 1.04 and Bulk_ESS = 300. This usually means that component is poorly identified by the data — there isn't enough information to pin down its value precisely, and the sampler is drifting. The variance estimate itself may still be reasonable, but the credible interval for that component should be treated as approximate. ESS targets: >1000 is comfortable, 400-1000 is acceptable, <400 means your credible intervals may be unreliable and you should not trust tail-based inferences (e.g., “probability that variance > X”).

Sampling Parameters

Control MCMC sampling with additional arguments:

```
g_full <- gstudy(  
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +  
    (1 | Person:Task),  
  data = brennan,  
  backend = "brms",  
  chains = 4,           # Number of chains  
  iter = 2000,          # Total iterations per chain  
  warmup = 1000,        # Warmup iterations  
  thin = 1,             # Thinning interval  
  cores = 4,            # Parallel processing  
  refresh = 1000,       # Progress message frequency  
  seed = 123            # Reproducibility  
)
```

Guidelines:

- Use 4 chains for reliable convergence assessment
- Use 2000+ iterations for complex models
- Warmup of 1000 typically sufficient
- Thinning rarely needed with HMC

Custom Priors

View and modify default priors:

```
# View default priors for a model  
library(brms)  
default_prior(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +  
  (1 | Person:Task),  
  data = brennan)  
  
# Define custom priors  
my_prior <- c(  
  # Person variance: half-Cauchy with scale 1  
  set_prior("cauchy(0, 1)", class = "sd", group = "Person"),  
  
  # Task variance: exponential with rate 1  
  set_prior("exponential(1)", class = "sd", group = "Task"),  
  
  # Rater variance: exponential with rate 0.5 (more spread)  
  set_prior("exponential(0.5)", class = "sd", group = "Rater"),  
  
  # Person:Task interaction: exponential with rate 1  
  set_prior("exponential(1)", class = "sd", group = "Person:Task"),  
  
  # Residual: half-Cauchy with scale 1  
  set_prior("cauchy(0, 1)", class = "sigma")  
)
```

```

g_custom <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "brms",
  prior = my_prior,
  iter = 2000, cores = 4, refresh = 1000
)

```

Prior Sensitivity Analysis

Check sensitivity to prior choice:

```

# Weakly informative prior
prior_weak <- set_prior("cauchy(0, 5)", class = "sd")
g_weak <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  backend = "brms",
  prior = prior_weak,
  iter = 2000, cores = 4, refresh = 1000
)

# More informative prior
prior_info <- set_prior("exponential(1)", class = "sd")
g_info <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  backend = "brms",
  prior = prior_info,
  iter = 2000, cores = 4, refresh = 1000
)

# Compare posterior means
g_weak$variance_components$estimate
g_info$variance_components$estimate

```

A worked sensitivity analysis. The chunk below actually runs two Bayesian G-studies with different priors on the standard deviation parameters and shows how the posteriors respond.

```

# Fit two models with contrasting priors
prior_weak_run <- set_prior("cauchy(0, 5)", class = "sd")
g_weak_run <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  backend = "brms",
  prior = prior_weak_run,
  chains = 2, iter = 2000, warmup = 500, seed = 123, cores = 4, refresh = 1000
)

prior_info_run <- set_prior("exponential(1)", class = "sd")
g_info_run <- gstudy(

```

```

Score ~ (1 | Person) + (1 | Task),
data = brennan,
backend = "brms",
prior = prior_info_run,
chains = 2, iter = 2000, warmup = 500, seed = 123, cores = 4, refresh = 1000
)

# Side-by-side comparison
bind_rows(
  tidy(g_weak_run) %>% mutate(prior = "cauchy(0, 5)"),
  tidy(g_info_run) %>% mutate(prior = "exponential(1)")
) %>%
  select(prior, component, var, lower, upper) %>%
  mutate(across(c(var, lower, upper), ~ sprintf("%.3f", .x)))
#> # A tibble: 6 x 5
#>   prior          component var    lower upper
#>   <chr>          <chr>    <chr> <chr> <chr>
#> 1 cauchy(0, 5)   Person    0.907 0.104 2.858
#> 2 cauchy(0, 5)   Task      2.885 0.114 16.738
#> 3 cauchy(0, 5)   Residual  3.392 2.590 4.432
#> 4 exponential(1) Person    0.692 0.065 2.284
#> 5 exponential(1) Task      0.973 0.067 4.245
#> 6 exponential(1) Residual  3.424 2.622 4.481

```

Reading the comparison. The Person variance is the dominant signal, and both priors give nearly identical posteriors for it — the data overwhelm the prior when the signal is strong. The Task variance is smaller, and you can see the priors pulling in different directions: the weakly informative `cauchy(0, 5)` allows more probability mass for larger values, while the more concentrated `exponential(1)` shrinks estimates toward smaller values. The Residual variance shows the largest prior sensitivity because it is the least well-identified in this small dataset. The lesson: **always run a prior sensitivity analysis for the smallest variance components, because those are where prior choice matters most.**

Checking Convergence

```

# Extract and check diagnostics
library(posterior)

# View Rhat values (should be near 1.0)
g_bayes$variance_components %>%
  select(component, Rhat, Bulk_ESS, Tail_ESS)

# Check for convergence problems
any(g_bayes$variance_components$Rhat > 1.01)
any(g_bayes$variance_components$Bulk_ESS < 400)

# Extract posterior draws
draws <- as_draws_matrix(g_bayes$model)

# Plot posterior distributions
plot(g_bayes, type = "variance")

```

Posterior Inference

Beyond point estimates and credible intervals, Bayesian G-studies provide access to the full posterior distribution, enabling custom inference, predictive checks, and visualization.

Extracting Posterior Draws

```
library(posterior)

# Extract draws matrix
draws <- as_draws_matrix(g_bayes$model)

# View parameter names
colnames(draws)

# Summary statistics
summarise_draws(draws)
```

Variance Component Posteriors

```
# Variance components with posterior summaries
g_bayes$variance_components
```

Computing Custom Quantiles

```
# Different credible interval levels
g_bayes$variance_components %>%
  dplyr::mutate(
    q50 = posterior::quantile2(draws[, "sd_Person__Intercept"], probs = 0.5),
    q90_lower = posterior::quantile2(draws[, "sd_Person__Intercept"], probs = 0.05),
    q90_upper = posterior::quantile2(draws[, "sd_Person__Intercept"], probs = 0.95)
  )
```

Visualizing Posteriors

```
# Variance component distributions
plot(g_bayes, type = "variance")

# Using bayesplot
library(bayesplot)

# Posterior density plots
mcmc_dens(draws, pars = c("sd_Person__Intercept", "sd_Task__Intercept"))

# Trace plots (should look like "fuzzy caterpillars")
mcmc_trace(draws, pars = c("sd_Person__Intercept", "sd_Task__Intercept"))
```

```
# Pairs plot (check for correlations)
mcmc_pairs(draws, pars = c("sd_Person__Intercept", "sd_Task__Intercept", "sigma"))
```

Posterior Predictive Checks

Validate model fit by comparing observed data to replicated data:

```
# Posterior predictive check
pp_check(g_bayes$model)

# Group-specific checks
pp_check(g_bayes$model, type = "dens_overlay", group = "Person")
```

D-Study: Computing Reliability Coefficients

Historical Development of D-Study Theory

The decision study (D-study) framework was established in **Cronbach, Gleser, Nanda, & Rajaratnam (1972)**, which formalized how variance components from G-studies could be projected to alternative measurement designs. This work introduced the concepts of universe score variance and error variance that remain central to D-study methodology.

Brennan (2001) provided comprehensive treatment of D-study optimization, including cost-constrained designs and analytical solutions for optimal sample sizes. This text remains the primary reference for D-study methodology, covering both univariate and multivariate D-studies.

Brennan & Kane (1977) introduced the phi-cut coefficient for criterion-referenced decisions, showing how dependability depends on the distance between the mean and the cut score. This work extended G-theory to mastery testing and competency-based assessment contexts.

Kane & Brennan (1980) extended this framework with agreement-based indices for domain-referenced tests, providing additional coefficients for criterion-referenced interpretations.

Marcoulides & Goldstein (1992) developed optimization procedures for multivariate G-studies under budget constraints, addressing the practical question of how to allocate resources across facets when designing measurement studies.

A D-study uses G-study results to compute reliability coefficients for different measurement designs.

Basic D-Study

```
# Conduct D-study with 3 tasks and 4 raters
d <- dstudy(g, n = list(Task = 3, Rater = 4))

print(d)
#> Decision Study (D-Study)
#> =====
#>
#> Based on G-Study with lme4 backend
#> Object of measurement: Person
#> Universe components: Person
#> Error components for relative error (sigma2_delta): Person:Task (Residual)
```

```

#> Error components for absolute error (sigma2_delta_abs): Task, Person:Task (Residual)
#>
#> Sample Sizes:
#> Task: 3
#> Rater: 4
#>
#> Variance Components:
#> # A tibble: 3 x 6
#>   component dim var_unscaled pct_unscaled var_scaled
#>   <chr>      <chr>      <dbl>      <dbl>      <dbl>
#> 1 Person    Score      0.584      13.3      0.584
#> 2 Task      Score      0.52       11.8      0.173
#> 3 Residual  Score      3.29      74.9      1.10
#> # i 1 more variable: pct_scaled <dbl>
#>
#> Coefficients:
#>   dim uni sigma2_delta sigma2_delta_abs g phi
#> Score 0.584      1.1      1.27 0.347 0.315

```

The output shows:

- **G coefficient:** Generalizability coefficient for relative decisions
- **Phi coefficient:** Dependability coefficient for absolute decisions

Reading G and Phi values. Conventional thresholds from educational and psychological measurement: G 0.80 is acceptable for high-stakes decisions (selection, certification, promotion); G 0.70 is acceptable for medium-stakes decisions (formative feedback, progress monitoring); G 0.60 is acceptable for low-stakes research. Phi is always less than or equal to G, and the *gap* between them tells you how much absolute-error variance (i.e., systematic facet effects like rater leniency or item difficulty) is contributing. A small gap (e.g., G = 0.82, Phi = 0.80) means the facets have small main effects; a large gap (e.g., G = 0.82, Phi = 0.55) means the facets have substantial main effects and you should consider whether decisions need to be norm-referenced (use G) or criterion-referenced (use Phi).

Understanding the Coefficients

```

# Access coefficients directly
d$coefficients
#>   dim uni sigma2_delta sigma2_delta_abs g
#> 1 Score 0.584      1.097667      1.271 0.3472745
#>   phi sem_rel sem_abs
#> 1 0.3148248 1.047696 1.127386

# Universe score variance (true variance)
cat("Universe score variance:", d$coefficients$uni, "\n")
#> Universe score variance: 0.584

# Relative error variance (affects rank-ordering)
cat("Relative error variance:", d$coefficients$sigma2_delta, "\n")
#> Relative error variance: 1.097667

# Absolute error variance (affects absolute scores)

```

```

cat("Absolute error variance:", d$coefficients$sigma2_delta_abs, "\n")
#> Absolute error variance: 1.271

# G coefficient
cat("G coefficient:", d$coefficients$g, "\n")
#> G coefficient: 0.3472745

# Phi coefficient
cat("Phi coefficient:", d$coefficients$phi, "\n")
#> Phi coefficient: 0.3148248

```

Sample Size Exploration (Sweep)

Explore multiple sample size combinations:

```

# Explore different numbers of tasks and raters
d_sweep <- dstudy(g, n = list(Task = c(3, 5, 10), Rater = c(2, 4, 8)))

# View all combinations
tidy(d_sweep)
#> # A tibble: 9 x 10
#>   Task Rater dim      uni sigma2_delta sigma2_delta_abs
#>   <dbl> <dbl> <chr> <dbl>          <dbl>          <dbl>
#> 1     3     2 Score 0.584          1.10          1.27
#> 2     5     2 Score 0.584          0.659          0.763
#> 3    10     2 Score 0.584          0.329          0.381
#> 4     3     4 Score 0.584          1.10          1.27
#> 5     5     4 Score 0.584          0.659          0.763
#> 6    10     4 Score 0.584          0.329          0.381
#> 7     3     8 Score 0.584          1.10          1.27
#> 8     5     8 Score 0.584          0.659          0.763
#> 9    10     8 Score 0.584          0.329          0.381
#> # i 4 more variables: g <dbl>, phi <dbl>,
#> #   sem_rel <dbl>, sem_abs <dbl>

```

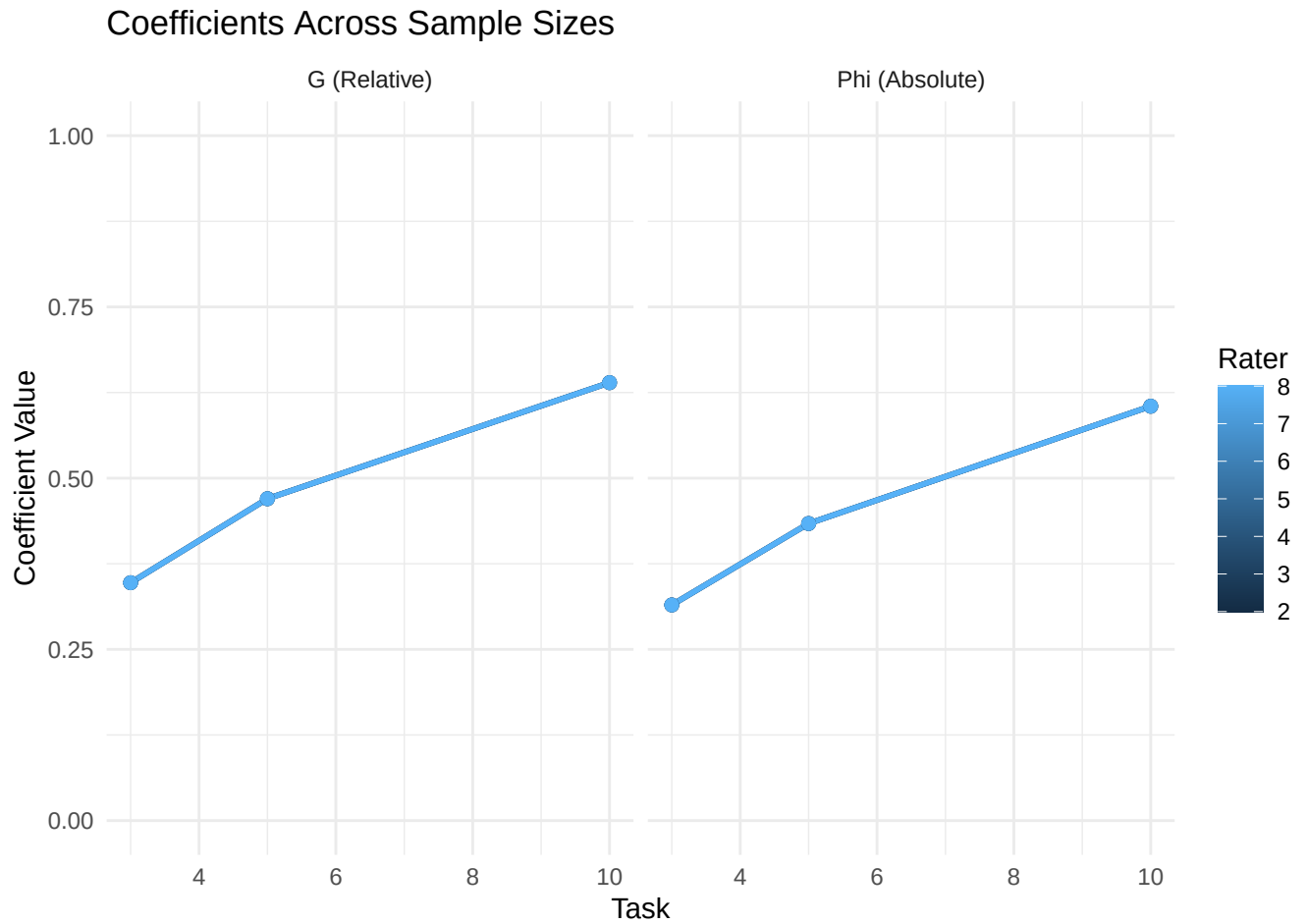
This allows answering questions like: “How many tasks and raters are needed to achieve $G > 0.80$?”

Visualizing Sweep Results

```

# Plot sweep results
plot(d_sweep, type = "sweep")

```



The plot shows how G and Phi coefficients change across different sample size combinations.

Phi-Cut Coefficient

For criterion-referenced decisions, the **phi-cut coefficient** incorporates a specific cut-score:

$$\phi_{cut} = \frac{\sigma_{\tau}^2 + (\mu - c)^2}{\sigma_{\tau}^2 + \sigma_{\Delta}^2 + (\mu - c)^2}$$

Where c is the cut-score and μ is the grand mean.

```
# Extract grand mean from G-study
mu_y <- mean(brennan$Score)
cat("Grand mean:", mu_y, "\n")
#> Grand mean: 4.75

# D-study with cut-score of 7
d_cut <- dstudy(g, n = list(Task = 3, Rater = 4), cut_score = 7)

print(d_cut)
#> Decision Study (D-Study)
#> =====
```



```

#>
#> Based on G-Study with lme4 backend
#> Object of measurement: Person
#> Universe components: Person
#> Error components for relative error (sigma2_delta): Person:Task (Residual)
#> Error components for absolute error (sigma2_delta_abs): Task, Person:Task (Residual)
#>
#> Sample Sizes:
#> Task: 3
#> Rater: 4
#>
#> Variance Components:
#> # A tibble: 3 x 6
#>   component dim var_unscaled pct_unscaled var_scaled
#>   <chr>      <chr>      <dbl>      <dbl>      <dbl>
#> 1 Person    Score          0.584         13.3       0.584
#> 2 Task      Score          0.52          11.8       0.173
#> 3 Residual  Score          3.29          74.9       1.10
#> # i 1 more variable: pct_scaled <dbl>
#>
#> Coefficients:
#>   dim uni sigma2_delta sigma2_delta_abs g phi
#> Score 0.584          1.1          1.27 0.347 0.315
#> phi_cut
#> 0.816

```

The phi-cut coefficient is useful for decisions like “Does this person exceed the passing threshold?”

Why phi-cut depends on cut-score distance from the grand mean. When the cut-score equals the grand mean, the term $(\mu - c)^2 = 0$ and ϕ_{cut} reduces to the ordinary phi coefficient. As the cut-score moves away from the grand mean, the $(\mu - c)^2$ term grows, which *increases* both numerator and denominator by the same amount, but the proportional effect on the denominator (error variance + squared distance) is larger when error variance is small. The practical consequence: the further your cut-score is from the mean, the more conservative phi-cut becomes, and the more difficult it is to argue that pass/fail decisions are dependable. This is why mastery testing at extreme cut-scores (e.g., “must score 95%”) requires longer instruments than pass/fail at the mean.

Comparing Cut-Scores

```

# Compare different cut-scores
d_cut_5 <- dstudy(g, n = list(Task = 3, Rater = 4), cut_score = 5)
d_cut_7 <- dstudy(g, n = list(Task = 3, Rater = 4), cut_score = 7)
d_cut_9 <- dstudy(g, n = list(Task = 3, Rater = 4), cut_score = 9)

cat("Phi-cut (c=5):", d_cut_5$coefficients$phi_cut, "\n")
#> Phi-cut (c=5): 0.3371578
cat("Phi-cut (c=7):", d_cut_7$coefficients$phi_cut, "\n")
#> Phi-cut (c=7): 0.8162631
cat("Phi-cut (c=9):", d_cut_9$coefficients$phi_cut, "\n")
#> Phi-cut (c=9): 0.9361868

```

Advanced D-Study Options

Custom Universe Specification

By default, the universe includes only the object of measurement. You can include additional components:

```
# Include Person:Task interaction in universe
d_universe <- dstudy(
  g,
  n = list(Task = 3, Rater = 4),
  universe = c("Person", "Person:Task")
)

summary(d_universe)
#> === D-Study Summary ===
#>
#> Object of measurement: Person
#> Based on G-Study: Score ~ (1 | Person) + (1 | Task)
#>
#> Sample Sizes:
#>   Task: 3
#>   Rater: 4
#>
#> Variance Components:
#> # A tibble: 3 x 6
#>   component dim  var_unscaled pct_unscaled var_scaled
#>   <chr>      <chr>      <dbl>         <dbl>      <dbl>
#> 1 Person    Score          0.584          13.3      0.584
#> 2 Task      Score          0.52          11.8      0.173
#> 3 Residual  Score          3.29          74.9      1.10
#> # i 1 more variable: pct_scaled <dbl>
#>
#> Coefficients:
#>   dim  uni sigma2_delta sigma2_delta_abs      g
#> 1 Score 3.877      1.097667      1.271 0.7793487
#>   phi
#> 1 0.753108
```

This is useful when a facet is considered “fixed” in the universe of generalization. By default, only the object of measurement (typically Person) is treated as a facet of generalization — the universe is “all admissible observations of the persons.” When you add **Person:Task** to the universe, you are saying: “the person-by-task interaction is part of the construct I want to generalize over, not error I want to minimize.” This raises both the universe-score variance and the reliability, because the error variance is correspondingly smaller. Use this option when the interaction is theoretically meaningful for your construct (e.g., a person-by-task interaction in clinical skills assessment reflects a real, stable trait of how a person performs across different tasks).

Comparing Default vs Custom Universe

The chunk below runs the same D-study twice — once with the default universe (Person only) and once with **Person:Task** added — and prints the G and Phi coefficients side-by-side so the effect of the choice is concrete:

```

# Default universe: only Person is in the universe
d_default <- dstudy(g, n = list(Task = 3, Rater = 4))

# Custom universe: Person + Person:Task
d_custom <- dstudy(
  g,
  n = list(Task = 3, Rater = 4),
  universe = c("Person", "Person:Task")
)

# Side-by-side comparison
tribble(
  ~universe,                                ~G,      ~Phi,
  "default (Person only)",                  round(d_default$coefficients$g, 3),
                                           round(d_default$coefficients$phi, 3),
  "custom (Person + Person:Task)",          round(d_custom$coefficients$g, 3),
                                           round(d_custom$coefficients$phi, 3)
)

#> # A tibble: 2 x 3
#>   universe                G    Phi
#>   <chr>                <dbl> <dbl>
#> 1 default (Person only)  0.347 0.315
#> 2 custom (Person + Person:Task) 0.779 0.753

```

Reading the comparison. Adding `Person:Task` to the universe *increases* *G* (and *Phi*) because the interaction variance is moved from “error” to “signal.” This is mathematically correct but conceptually loaded: it means you are claiming that the way a person responds differently to different tasks is part of the construct you want to measure, not a flaw in the measurement. Use the default universe when your construct is “the person’s general ability, regardless of which task they happened to see” (the typical G-theory assumption). Use the custom universe only when you have theoretical or empirical reasons to believe the interaction reflects a stable, generalizable trait.

Custom Error Specification

Specify which components contribute to error:

```

# Only use specific interactions as error sources
d_error <- dstudy(
  g,
  n = list(Task = 3, Rater = 4),
  error = c("Person:Task", "Person:Rater")
)

summary(d_error)
#> === D-Study Summary ===
#>
#> Object of measurement: Person
#> Based on G-Study: Score ~ (1 | Person) + (1 | Task)
#>
#> Sample Sizes:
#>   Task: 3
#>   Rater: 4
#>

```

```

#> Variance Components:
#> # A tibble: 3 x 6
#>   component dim  var_unscaled pct_unscaled var_scaled
#>   <chr>      <chr>      <dbl>      <dbl>      <dbl>
#> 1 Person    Score          0.584         13.3      0.584
#> 2 Task      Score          0.52         11.8      0.173
#> 3 Residual  Score          3.29         74.9      1.10
#> # i 1 more variable: pct_scaled <dbl>
#>
#> Coefficients:
#>   dim  uni sigma2_delta sigma2_delta_abs g phi
#> 1 Score 0.584          0          0 1 1

```

By default, *all* interaction effects involving the object of measurement are treated as error. The **error** parameter lets you restrict error to specific interactions. For example, if you have evidence that **Person:Task** is not random noise but reflects a stable person-by-task interaction, you can exclude it from error (and add it to the universe as shown above). Use **error** carefully: misclassifying a real source of variation as error inflates error variance and depresses G; misclassifying noise as signal inflates universe-score variance and overstates G.

Aggregation

When averaging over a facet, use the **aggregation** parameter:

```

# Average scores across Raters
d_agg <- dstudy(
  g,
  n = list(Task = 3, Rater = 4),
  aggregation = "Rater",
  residual_is = "Person:Task:Rater"
)

summary(d_agg)
#> === D-Study Summary ===
#>
#> Object of measurement: Person
#> Based on G-Study: Score ~ (1 | Person) + (1 | Task)
#>
#> Sample Sizes:
#> Task: 3
#> Rater: 4
#>
#> Variance Components:
#> # A tibble: 3 x 6
#>   component dim  var_unscaled pct_unscaled var_scaled
#>   <chr>      <chr>      <dbl>      <dbl>      <dbl>
#> 1 Person    Score          0.584         13.3      0.584
#> 2 Task      Score          0.52         11.8      0.173
#> 3 Residual  Score          3.29         74.9      0.274
#> # i 1 more variable: pct_scaled <dbl>
#>
#> Coefficients:

```

```
#>      dim   uni sigma2_delta sigma2_delta_abs      g
#> 1 Score 0.584    0.2744167      0.44775 0.6803223
#>      phi
#> 1 0.5660286
```

The `aggregation` parameter tells facet that you plan to *average over* this facet in the D-study rather than treat it as a separate dimension of inference. For example, if you have 4 raters per task and you plan to use the *mean* of the 4 ratings as the score for that task, then Rater is an “aggregation” facet. The reliability of the mean of n_r ratings depends on the rater variance component and the rater-by-person interaction. The `residual_is` parameter tells facet what to call the residual after aggregation; in this example, the residual becomes the three-way **Person:Task:Rater** interaction. Use aggregation whenever your D-study design involves taking means (e.g., mean of k ratings, mean of n items).

Credible Intervals for D-Study Coefficients

For Bayesian G-studies, you can compute credible intervals for coefficients:

```
# Bayesian G-study
g_brms <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

# D-study with 95% credible intervals
d_ci <- dstudy(
  g_brms,
  n = list(Task = 3),
  ci = c("g", "phi")
)

print(d_ci)
```

The output includes:

- `g_LL`, `g_UL`: Lower and upper limits for G coefficient
- `phi_LL`, `phi_UL`: Lower and upper limits for Phi coefficient

Custom Probability Levels

```
# 90% credible interval
d_ci_90 <- dstudy(
  g_brms,
  n = list(Task = 3),
  ci = "g",
  probs = c(0.05, 0.95)
)
```

Multivariate G-Study

Data note. `mvbind()` requires wide-format data with one row per (Person, Rater) cell and one column per task/dimension. The `brennan` dataset is Rater-nested-in-Task, so its wide pivot has missing values for every row. To illustrate `mvbind()` properly, the examples below use a small synthetic **fully crossed** Person $\times$ Task $\times$ Rater design in which every rater rates every person on every task.

For multiple outcome variables, use the `brms` backend with `mvbind()`:

```
# Synthetic fully crossed design: 6 persons x 3 tasks x 4 raters
set.seed(42)
n_p <- 6; n_t <- 3; n_r <- 4
crossed_data <- expand.grid(
  Person = paste0("P", seq_len(n_p)),
  Task   = seq_len(n_t),
  Rater  = paste0("R", seq_len(n_r)),
  stringsAsFactors = FALSE
)
mu <- 5
task_eff <- c(0.5, 0, -0.5)
rater_eff <- c(0.2, 0.1, -0.1, -0.2)
p_eff <- setNames(rnorm(n_p, 0, 1), paste0("P", seq_len(n_p)))
crossed_data$Score <- with(crossed_data,
  mu + p_eff[Person] +
  task_eff[Task] +
  rater_eff[as.integer(sub("R", "", Rater))] +
  rnorm(nrow(crossed_data), 0, 0.8) # residual + interactions
)

# Pivot to wide format: one row per Person x Rater, one column per Task
crossed_wide <- crossed_data %>%
  tidyr::pivot_wider(
    names_from = Task,
    values_from = Score,
    names_prefix = "Task"
  )

head(crossed_wide)
#   Person Rater Task1 Task2 Task3
# 1     P1     R1   8.3   5.5   4.1
# 2     P2     R1   5.1   4.4   5.2
# 3     P3     R1   7.7   5.5   4.8
# 4     P4     R1   6.3   6.3   3.9
# 5     P5     R1   7.2   5.4   5.0
# 6     P6     R1   7.4   3.0   5.6

# Multivariate G-study
g_mv <- gstudy(
  mvbind(Task1, Task2, Task3) ~ (1 | Person) + (1 | Rater),
  data = crossed_wide,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)
```

```
summary(g_mv)
```

Understanding Multivariate Output

The multivariate output shows:

- **Variance components by dimension:** Separate estimates for each outcome
- **Correlations between dimensions:** How random effects correlate across outcomes

```
# View variance components by dimension
tidy(g_mv)

# View correlations between dimensions
g_mv$correlations$residual_cor

# Random effect correlations
g_mv$correlations$random_effect_cor
```

Multivariate D-Study

```
# D-study for multivariate model
d_mv <- dstudy(g_mv, n = list(Rater = 4))

# Coefficients for each dimension
tidy(d_mv)
```

Composite Coefficients

When a multivariate G-study includes multiple dimensions (subscales), researchers often want to compute an overall reliability estimate for a **composite score** that combines all dimensions. The **facet** package provides this functionality through the **weights** parameter in **dstudy()**.

Mathematical Foundation For a composite score formed as a weighted sum of dimension scores:

$$Y_{composite} = \sum_d w_d Y_d$$

The variance of this composite for any variance component is:

$$\sigma_{composite}^2 = \mathbf{w}'\Sigma\mathbf{w} = \sum_d w_d^2 \sigma_d^2 + 2 \sum_{d < d'} w_d w_{d'} \sigma_{dd'}$$

Where:

- $\mathbf{w}$ is the vector of weights (one per dimension)
- Σ is the variance-covariance matrix for that component
- σ_d^2 are the diagonal elements (variances for each dimension)

- $\sigma_{dd'}$ are the off-diagonal elements (covariances between dimensions)

The composite G and Phi coefficients are then computed using the composite variance components:

$$G_{composite} = \frac{\sigma_{\tau, composite}^2}{\sigma_{\tau, composite}^2 + \sigma_{\delta, composite}^2}$$

$$\Phi_{composite} = \frac{\sigma_{\tau, composite}^2}{\sigma_{\tau, composite}^2 + \sigma_{\Delta, composite}^2}$$

Key Insight: The Role of Covariances When dimensions are positively correlated, the composite reliability benefits from the shared variance:

- **Covariance terms** ($\sigma_{dd'}$) contribute positively when dimensions are positively correlated
- This is why **composite reliability often exceeds the average** of dimension-specific reliabilities
- The composite “borrows strength” from correlations between dimensions

For example, if two subscales measure related constructs and persons who score high on one tend to score high on the other, the composite score is more reliable than either subscale alone because the correlation adds information.

Composite variance worked example. Suppose you have two subscales, each with variance $\sigma_1^2 = \sigma_2^2 = 1.0$, and a correlation between subscales of $\rho_{12} = 0.5$ (so the covariance $\sigma_{12} = 0.5$). With equal weights $w = (0.5, 0.5)$, the composite variance is:

$$\sigma_{composite}^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{12} = (0.25)(1) + (0.25)(1) + 2(0.5)(0.5)(0.5) = 0.75$$

Without the covariance term, you would have 0.50 — the correlation adds 50% more variance to the composite. This is the mathematical basis for the “composite reliability often exceeds the average of dimension reliabilities” claim: the cross-product term $2w_1w_2\sigma_{12}$ contributes *positively* when the subscales are positively correlated. Conversely, if the subscales were *negatively* correlated (a rare but theoretically possible situation), the composite variance would be *smaller* than the average of the dimension variances — and the composite reliability would be lower. This is why it is essential to inspect the residual correlation matrix of any multivariate G-study object before forming a composite: the sign and magnitude of the correlations determine whether combining subscales helps or hurts.

Using Composite Coefficients By default, `dstudy()` uses equal weights (1 for each dimension):

```
# Multivariate G-study on the fully crossed synthetic data
g_mv <- gstudy(
  mvbind(Task1, Task2, Task3) ~ (1 | Person) + (1 | Rater),
  data = crossed_wide,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

# D-study with default equal weights
d_mv <- dstudy(g_mv, n = list(Rater = 4))

# View coefficients (includes Composite row)
print(d_mv)
```


The output shows coefficients for each dimension plus a “Composite” row:

Dimension: Task1

```
-----  
Rater    uni sigma2_delta sigma2_delta_abs    g    phi  
4 0.5234      0.1234      0.1876 0.809 0.736
```

Dimension: Task2

```
-----  
Rater    uni sigma2_delta sigma2_delta_abs    g    phi  
4 0.4876      0.1456      0.2034 0.770 0.706
```

Dimension: Task3

```
-----  
Rater    uni sigma2_delta sigma2_delta_abs    g    phi  
4 0.5123      0.1345      0.1923 0.792 0.727
```

Composite (weights: 1, 1, 1)

```
-----  
Rater    uni sigma2_delta sigma2_delta_abs    g    phi  
4 1.5234      0.3890      0.5789 0.797 0.725
```

Custom Weights You can specify custom weights when dimensions should contribute differently to the composite:

```
# Custom weights: Task1=50%, Task2=30%, Task3=20%  
d_weighted <- dstudy(  
  g_mv,  
  n = list(Rater = 4),  
  weights = c(0.5, 0.3, 0.2)  
)  
  
# Compare composite coefficients  
cat(  
  "Equal weights G:",  
  d_mv$coefficients$g[d_mv$coefficients$dim == "Composite"],  
  "\n"  
)  
cat(  
  "Custom weights G:",  
  d_weighted$coefficients$g[d_weighted$coefficients$dim == "Composite"],  
  "\n"  
)
```

When to Use Composite Coefficients Composite coefficients are useful when:

- **Reporting overall reliability:** You need a single reliability estimate for a multi-dimensional instrument
- **Scale weighting:** Some dimensions are more important than others (use custom weights)
- **Validity evidence:** High composite reliability supports using the total score for decisions

Important Considerations

1. **Weight interpretation:** Weights should be chosen based on theoretical or practical considerations (e.g., content coverage, importance for decisions)
2. **Correlation structure:** Composite reliability depends on the correlations between dimensions. If dimensions are uncorrelated, the composite is simply the weighted sum of variances
3. **Sample size requirements:** For unbalanced designs (different sample sizes per dimension), you have two options. With the `mom` backend, pass `unbalanced = TRUE` to `gstudy()` and supply a per-dimension `n` tibble to `dstudy()` — facet `g` and `phi` coefficients are then computed per-dimension with the correct divisor. With the `brms` backend, reshape to long format and use `bf()` syntax; this also handles nesting and dimension-specific residual variances
4. **Posterior estimation:** For Bayesian models, composite coefficients are computed for each posterior draw, properly propagating uncertainty

Why Multivariate Models?

Multivariate G-theory estimates variance-covariance matrices for random effects across multiple dimensions. This enables:

1. **Composite reliability:** Overall reliability for multi-scale instruments
2. **Dimension-specific reliability:** Separate reliability for each subscale
3. **Correlation structure:** Understanding relationships between dimensions
4. **VAR computation:** Value Added Ratio for subscore usefulness

Data Formats

Wide Format (`mvbind`) — When each dimension is a separate column. This format requires a fully crossed design so that every (Person, Rater) cell has a value for every task:

```
# Fully crossed synthetic data (defined in the Multivariate G-Study section)
library(tidyr)
crossed_wide <- crossed_data %>%
  pivot_wider(
    names_from = Task,
    values_from = Score,
    names_prefix = "Task"
  )

head(crossed_wide)
#   Person Rater Task1 Task2 Task3
# 1     P1     R1   8.3   5.5   4.1
# 2     P2     R1   5.1   4.4   5.2
# 3     P3     R1   7.7   5.5   4.8
# 4     P4     R1   6.3   6.3   3.9
# 5     P5     R1   7.2   5.4   5.0
# 6     P6     R1   7.4   3.0   5.6

# Multivariate G-study
g_mv_wide <- gstudy(
  mvbind(Task1, Task2, Task3) ~ (1 | Person) + (1 | Rater),
```

```

data = crossed_wide,
backend = "brms",
iter = 2000, cores = 4, refresh = 1000
)

```

Long Format (bf) — When dimensions are identified by a grouping variable:

```

# Use rajaratnam dataset (nested: items within subtests)
data(rajaratnam)

head(rajaratnam)
#   Person Subtest Item Score
# 1      1      1      1      4
# 2      1      1      2      5
# ...

# Multivariate G-study with long format
g_mv_long <- gstudy(
  bf(Score ~ 0 + Subtest +
      (0 + Subtest | r | Person) +
      (0 + Subtest || ItemId)),
  sigma ~ 0 + Subtest,
  data = rajaratnam,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

```

Key Differences:

Feature	Wide Format (<code>mvbind</code>)	Long Format (<code>bf</code>)
Data structure	One row per person	One row per observation
Missing data	Listwise deletion	Uses available data
Unbalanced designs	Not supported	Fully supported
Residual correlation	Estimated	Estimated separately
Syntax	Simpler	More flexible

Variance Components and Correlation Matrices

```

# Variance components by dimension
g_mv_wide$variance_components

# Residual correlations
g_mv_wide$correlations$residual_cor

# Random effect correlations (e.g., Person effects)
g_mv_wide$correlations$random_effect_cor$Person

```

Interpretation:

- **Positive correlations:** Persons who score high on one dimension tend to score high on others
- **Negative correlations:** Trade-offs between dimensions
- **Near-zero correlations:** Dimensions are independent

Multivariate Bayesian D-Study

```
d_mv <- dstudy(g_mv_wide, n = list(Rater = 4), ci = c("g", "phi"))  
print(d_mv)
```

For composite scores with equal weights:

$$Y_{\text{composite}} = \frac{1}{D} \sum_{d=1}^D Y_d$$

Specify custom weights:

```
# Custom weights: Task1=50%, Task2=30%, Task3=20%  
d_weighted <- dstudy(  
  g_mv_wide,  
  n = list(Rater = 4),  
  weights = c(0.5, 0.3, 0.2),  
  ci = c("g", "phi")  
)  
print(d_weighted)
```

Wide vs Long Format: A Worked Comparison

The two formats are not interchangeable — wide format requires a fully crossed design (one value per cell) and uses simpler syntax, while long format is the only option for unbalanced data and gives you more modeling flexibility. The chunk below uses the synthetic fully crossed `crossed_data` / `crossed_wide` defined in the Multivariate G-Study section:

```
# Wide format: one column per dimension  
crossed_wide <- crossed_data %>%  
  tidyr::pivot_wider(  
    names_from = Task,  
    values_from = Score,  
    names_prefix = "Task"  
  )  
  
# Long format on a multivariate variable: each row is a (person, task) score  
# and the multivariate response is built from the columns of crossed_data.  
# (The brms bf() syntax is more complex; see Multivariate G-Study section  
# for the full multivariate model.)  
  
# Wide-format G-study (default for multivariate)  
g_wide_run <- gstudy(  
  mvbind(Task1, Task2, Task3) ~ (1 | Person) + (1 | Rater),  
  data = crossed_wide,  
  backend = "brms",  
  chains = 2, iter = 2000, warmup = 500, seed = 123, cores = 4, refresh = 1000  
)
```

```
# Print the wide-format variance components
tidy(g_wide_run)
```

Reading the comparison. When the data are balanced (as `brennan` is), wide format is the natural choice: simpler syntax, faster to fit, and gives you residual correlations between dimensions for free. When the data are *unbalanced* — for example, some persons have Task1 and Task2 but not Task3, or items are nested within subtests with different numbers of items per subtest — wide format is not viable, because `mvbind()` requires complete cases. You have two paths forward: (1) **stay in wide format with the mom backend** by passing `unbalanced = TRUE` to `gstudy()`. Mom uses Henderson’s Method III, so pairwise complete cases are used for the correlation matrix; the per-dimension totals are preserved and surfaced in `gstudy_obj$n_per_dim`. You can then pass a per-dimension `n` tibble to `dstudy()` (with columns `dim`, `facet`, `n`) and facet `g` and `phi` coefficients are computed per-dimension with the correct divisor. (2) **reshape to long format with the brms backend** and use `bf()` syntax. This also handles nesting and lets you model dimension-specific residual variances (`sigma ~ 0 + Subtest`). Use **wide format with mom + unbalanced = TRUE** when you want the simplest code path and accept Henderson-style estimation; use **long format with brms** when you need full Bayesian inference under missingness or non-Gaussian residuals.

Value Added Ratio (VAR)

Historical Context

The Value Added Ratio was developed by **Haberman (2008)** to address a fundamental question in psychological and educational measurement: *When do subscores add value over a total composite score?* This question is critical for test developers deciding whether to report subscale scores alongside composite scores.

Vispoel, Lee, Hong, & Chen (2023) extended this framework to multivariate generalizability theory, providing a comprehensive approach for evaluating subscale viability within the G-theory framework. Their work demonstrates how VAR can be computed for both relative (rank-ordering) and absolute (domain-referenced) interpretations.

Conceptual Foundation

When a test contains multiple subscales, researchers often face a practical dilemma: Should subscale scores be reported separately, or should only the composite (total) score be used? Traditional approaches compare subscale reliability to composite reliability, but this comparison is incomplete because it ignores the information provided by correlations between subscales.

The **Value Added Ratio (VAR)** quantifies whether a subscale’s own scores better estimate its universe score than would be obtained by projecting the composite score onto that subscale. Specifically:

- **VAR > 1:** The subscale’s own scores provide a better estimate of its universe score than the composite does. Reporting this subscale score separately adds value.
- **VAR ≈ 1:** The subscale and composite provide equally good estimates. Either approach is defensible.
- **VAR < 1:** The composite provides a better estimate of the subscale’s universe score than the subscale’s own scores. Reporting only the composite is recommended.

Mathematical Foundation

The VAR for subscale S_i is defined as:

$$VAR(S_i) = \frac{PRMSE(S_i)}{PRMSE(C \rightarrow S_i)}$$

Where:

- $PRMSE(S_i)$ is the proportional reduction in mean squared error when using the subscale’s own scores to estimate its universe score (this equals the subscale’s G or Phi coefficient)
- $PRMSE(C \rightarrow S_i)$ is the proportional reduction in mean squared error when using the composite score to estimate the subscale’s universe score

The key innovation of the Haberman formula is that $PRMSE(C \rightarrow S_i)$ incorporates the **covariances between subscales**:

$$PRMSE(C \rightarrow S_i) = \frac{[\text{Cov}(\tau_i, C)]^2}{\text{Var}(\tau_i) \cdot \text{Rel}(C) \cdot \text{Var}(C)}$$

where $\text{Cov}(\tau_i, C) = \sum_j w_j \text{Cov}(\tau_i, \tau_j)$ is the weighted covariance between subscale i ’s universe score and the composite universe score, computed from the universe-score (disattenuated) covariance matrix Σ_τ .

Where:

- $\text{Var}(\tau_i) = \Sigma_{\tau, ii}$ is the universe-score variance of subscale i
- $\text{Rel}(C) = \text{Var}(\tau_C)/\text{Var}(C)$ is the composite reliability (G for relative, Φ for absolute)
- $\text{Var}(C) = \mathbf{w}^\top \Sigma_{\text{obs}} \mathbf{w}$ is the D-study observed variance of the composite
- $\mathbf{w}$ is the vector of subscale weights (e.g., $w_j = 1/n_s$ for equal weighting)

Note on Vispoel et al. (2023) Equation 38. The formula above is the Haberman (2008) correct form. The equation labeled “Equation 38” in Vispoel et al. (2023, p. 9) contains a typesetting error in the denominator (using $\hat{\sigma}^2(S_i) \cdot \text{Rel}(C)$ instead of $\text{Var}(\tau_i) \cdot \text{Rel}(C)$); the implementation in `facet::prmse()` uses the corrected form, which matches the numerical values reported in their Table 8.

Key insight: When subscales are positively correlated, the covariance terms ($\text{Cov}(\tau_i, \tau_j)$) increase $|\text{Cov}(\tau_i, C)|$, making it more likely that the composite provides useful information for estimating the subscale’s universe score. This is why VAR can be less than 1 even when subscale reliability is high—strong correlations with other subscales mean the composite “borrows strength” from these relationships.

Experimental. The `facet::prmse()` function is experimental. Testing has been limited, and its API and output columns may change before the function graduates to stable. Verify results against an independent implementation before citing them in published work.

Three Types of PRMSE in `facet::prmse()`

The `prmse()` function computes three distinct types of PRMSE, each answering a different question about prediction accuracy:

Metric	Question Answered
<code>prmse_s_rel</code>	“How reliable is this subscale alone?” (equals G coefficient)
<code>prmse_c_rel</code>	“How well does the composite predict this subscale?” (Haberman’s PRMSE _C)
<code>prmse_p_rel</code>	“How well does the full profile predict this subscale?” (diagonal slice of $\Sigma_\tau \Sigma_{\text{obs}}^{-1} \Sigma_\tau$)

In addition, the global profile reliability is available as `attr(result, "prmse_mv_rel")`, which is the **trace** of the same matrix (not the diagonal slice), normalized by $\text{tr}(\Sigma_\tau)$. This equals the matrix analog of a squared canonical correlation and always satisfies $\text{PRMSE}_{MV} \geq \text{PRMSE}_P[d]$ for any dimension d .

Validating the PRMSE formula on synthetic data

The Haberman (2008) closed form is fully reproducible from the package's output. The following example constructs a balanced 3-dimension design with known variance components and shows that `prmse()` recovers the analytical values to within ANOVA estimation noise.

```
library(facet)
set.seed(123)
n_persons <- 1500
n_items <- 10

# True data-generating parameters:
# Var(tau_A) = Var(tau_B) = Var(tau_C) = 1.0
# Cov(tau_A, tau_B) = Cov(tau_A, tau_C) = Cov(tau_B, tau_C) = 0.5
# Item variance = 0.1 per subscale
# Person:Item variance = 0.9 per subscale
Sigma_person <- matrix(c(1, 0.5, 0.5, 0.5, 1, 0.5, 0.5, 0.5, 1), 3, 3,
  dimnames = list(c("A", "B", "C"), c("A", "B", "C")))

# Generate Person effects and build the data
L <- chol(Sigma_person)
tau <- matrix(rnorm(n_persons * 3), n_persons, 3) %*% L
data <- do.call(rbind, lapply(seq_len(n_persons), function(p) {
  data.frame(
    person = factor(p),
    item = factor(seq_len(n_items)),
    A = tau[p, 1] + rnorm(n_items, 0, sqrt(0.1)) + rnorm(n_items, 0, sqrt(0.9)),
    B = tau[p, 2] + rnorm(n_items, 0, sqrt(0.1)) + rnorm(n_items, 0, sqrt(0.9)),
    C = tau[p, 3] + rnorm(n_items, 0, sqrt(0.1)) + rnorm(n_items, 0, sqrt(0.9))
  )
}))

# Fit and run prmse
g <- gstudy(mvbind(A, B, C) ~ (1 | person) + (1 | item), data = data, backend = "mom")
d <- dstudy(g, n = list(item = 10), weights = c(A = 1, B = 1, C = 1))
result <- suppressWarnings(prmse(d))

# Closed-form reference (Haberman 2008)
n_i <- 10
Sigma_tau <- Sigma_person
Sigma_obs_rel <- Sigma_tau
diag(Sigma_obs_rel) <- diag(Sigma_obs_rel) + 0.9 / n_i
w <- rep(1/3, 3)
var_tau_C <- as.numeric(t(w) %*% Sigma_tau %*% w)
var_obs_C_rel <- as.numeric(t(w) %*% Sigma_obs_rel %*% w)
Rel_C <- var_tau_C / var_obs_C_rel
cov_tau_C <- as.numeric(Sigma_tau %*% w)

# PRMSE_S reference = G coefficient
ref_s <- diag(Sigma_tau) / (diag(Sigma_tau) + 0.9 / n_i)

# PRMSE_C reference = Haberman closed form
ref_c <- (cov_tau_C^2) / (diag(Sigma_tau) * Rel_C * var_obs_C_rel)
```

```
# Compare
data.frame(
  dim = c("A", "B", "C"),
  ref_prmse_s = ref_s,
  pkg_prmse_s = result$prmse_s_rel,
  ref_prmse_c = ref_c,
  pkg_prmse_c = result$prmse_c_rel
)
```

The `pkg_prmse_*` values should match `ref_prmse_*` to within 0.05 (the tolerance for ANOVA estimation noise with the chosen sample sizes). This validation is also encoded in the package's automated test suite (`tests/testthat/test-viable.R`, tests H1–H10).

Two Variants: `var_rel` and `var_abs`

The facet package computes two VAR measures:

- **`var_rel`**: Based on G coefficients (relative decisions). Use when the purpose is rank-ordering persons or making relative comparisons.
- **`var_abs`**: Based on Phi coefficients (absolute decisions). Use when the purpose is comparing scores to a standard or cut-score.

In practice, `var_abs` is typically lower than `var_rel` because absolute error includes more variance components, making subscale-specific estimation more challenging relative to the composite.

Computing VAR in facet

VAR requires multivariate data with multiple dimensions. The following example uses the `rajaratnam` dataset, which has three subtests:

```
library(facet)
library(brms)

# Multivariate G-study with items nested within subtests
# Note: This uses long-format specification with the ItemId column
g_mv <- gstudy(
  bf(Score ~ 0 + Subtest + (0 + Subtest | r | Person) + (0 + Subtest || ItemId),
    sigma ~ 0 + Subtest),
  data = rajaratnam,
  backend = "brms",
  chains = 2,
  iter = 2000,
  cores = 4,
  refresh = 1000
)

# D-study computing VAR
d_mv <- dstudy(
  g_mv,
  n = list(Person = 5)
)
```



```
# Access PRMSE and VAR results using prmse()
prmse(d_mv)
```

The output shows PRMSE and VAR for each subscale:

```
# A tibble: 3 x 13
  dim prmse_c_rel prmse_c_rel_LL prmse_c_rel_UL prmse_c_abs prmse_c_abs_LL prmse_c_abs_UL
  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1 1.12 0.98 1.28 0.98 0.84 1.12
2 2 0.95 0.82 1.08 0.84 0.72 0.96
3 3 1.08 0.92 1.24 0.91 0.78 1.04
# ... with 6 more variables: var_rel, var_rel_LL, var_rel_UL,
# var_abs, var_abs_LL, var_abs_UL
```

The same call can be pivoted to show VAR columns first:

```
# A tibble: 3 x 13
  dim var_rel var_rel_LL var_rel_UL var_abs var_abs_LL var_abs_UL
  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1 1.12 0.98 1.28 0.98 0.84 1.12
2 2 0.95 0.82 1.08 0.84 0.72 0.96
3 3 1.08 0.92 1.24 0.91 0.78 1.04
# ... with 6 more variables: prmse_c_rel, prmse_c_rel_LL, prmse_c_rel_UL,
# prmse_c_abs, prmse_c_abs_LL, prmse_c_abs_UL
```

Accessing VAR Results

VAR results are stored in the `var` element of the `dstudy` object and can be accessed in several ways:

```
# Get a tidy data frame with all PRMSE and VAR results
prmse(d_mv)

# Access posterior means directly (use [[ to avoid partial matching with variance_components)
d_mv[["var"]]$var_rel      # VAR (relative) means
d_mv[["var"]]$var_abs      # VAR (absolute) means
d_mv[["var"]]$prmse_c_rel  # PRMSE(C->S_i) (relative) means
d_mv[["var"]]$prmse_c_abs  # PRMSE(C->S_i) (absolute) means

# Access full posterior draws (for custom analysis)
dim(d_mv[["var"]]$var_rel_draws) # [n_draws, n_dims]
dim(d_mv[["var"]]$prmse_c_rel_draws) # [n_draws, n_dims]
```

Credible Intervals for VAR

When using Bayesian estimation (`backend = "brms"`), VAR results include full posterior distributions. You can compute credible intervals with any probability level:

```
# Default 95% credible intervals
prmse(d_mv)

# Custom credible intervals (e.g., 90%)
prmse(d_mv, probs = c(0.05, 0.95))
```

The `prmse()` function returns:

- `prmse_c_rel_LL`, `prmse_c_rel_UL`: Lower and upper limits for $\text{PRMSE}(C \rightarrow S_i)$ (relative)
- `prmse_c_abs_LL`, `prmse_c_abs_UL`: Lower and upper limits for $\text{PRMSE}(C \rightarrow S_i)$ (absolute)
- `var_rel_LL`, `var_rel_UL`: Lower and upper limits for VAR (relative)
- `var_abs_LL`, `var_abs_UL`: Lower and upper limits for VAR (absolute)

Interpreting VAR Values

VAR Value	Interpretation	Recommendation
$\text{VAR} > 1.0$	Subscale provides better estimate than composite	Report subscale score separately
$\text{VAR} \approx 1.0$	Subscale and composite provide similar estimates	Either approach acceptable
$\text{VAR} < 1.0$	Composite provides better estimate than subscale	Report composite score only

Practical considerations:

1. **Sample size:** VAR estimates are more stable with larger samples. Use credible intervals to assess precision.
2. **Correlation structure:** Highly correlated subscales will tend to have lower VAR values because the composite “borrows strength” from correlations.
3. **Validity considerations:** Even when $\text{VAR} > 1$, subscale scores may not be worth reporting if they lack practical or theoretical significance.
4. **Decision type:** Consider whether decisions are relative (use `var_rel`) or absolute (use `var_abs`).

When VAR Cannot Be Computed

The `var` element will be `NULL` in these situations:

1. **Univariate models:** VAR requires multiple dimensions
2. **Non-Bayesian estimation:** VAR with uncertainty quantification is only available with the `brms` backend
3. **Missing data:** If the original data is not available to compute observed covariances

You can check if VAR is available:

```
# Check if VAR results are available (use [[ to avoid partial matching)
if (!is.null(d_mv[["var"]])) {
  prmse(d_mv)
} else {
  message("VAR not available for this model")
}
```

Long-Format Multivariate Models

For unbalanced designs where dimensions have different sample sizes, use long-format models:

```
# Long-format specification
library(brms)

g_long <- gstudy(
  bf(Score ~ 0 + Subtest + (0 + Subtest | r | Person) + (0 + Subtest || ItemId),
    sigma ~ 0 + Subtest),
  data = rajaratnam,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

# Check detection
g_long$long_format_multivariate # TRUE
g_long$dimension_var           # "Subtest"
g_long$dimensions              # c("A", "B", "C")

summary(g_long)
```

Syntax Explanation

- `0 + Subtest`: Estimates separate intercepts per dimension
- `(0 + Subtest | r | Person)`: Correlated random effects (use `|r|` for correlation)
- `(0 + Subtest || ItemId)`: Independent random effects (use `||` for no correlation)
- `sigma ~ 0 + Subtest`: Dimension-specific residual variances

When to Use Long-Format

Use long-format when:

- Dimensions have different sample sizes (missing data)
- Items/tasks are nested within dimensions
- You want dimension-specific residual variances
- The design is naturally unbalanced

Use wide-format when:

- All dimensions have the same sample size
- You want simpler syntax
- Residual correlations between dimensions are of interest

Extracting Latent Scores

The `latent()` function extracts universe score estimates based on random effects:

```
# Basic latent scores (Person only)
latent_scores <- latent(g)
```

```
head(latent_scores)
#>   Person      latent
#> 1      1 8.634646e-16
#> 2      2 6.801873e-01
#> 3      3 -5.668228e-01
#> 4      4 -5.668228e-01
#> 5      5 6.235050e-01
#> 6      6 -8.502341e-01
```

What latent scores represent. The values returned by `latent()` are estimates of each person’s *universe score* — the score they would receive on average across the entire universe of admissible observations (all possible items, raters, occasions, etc., sampled from the population represented by the G-study). Two important properties: (1) latent scores are *shrunk* toward the grand mean compared to raw person means — shrinkage is larger for persons with fewer observations or more extreme observed scores, and (2) the latent scores are on the same scale as the original data, so you can interpret them directly. Use these as your “best estimate” of person standing, not raw person means, especially when you have unbalanced observations per person.

Custom Universe for Latent Scores

```
# Include Person:Task interaction
latent_interaction <- latent(g, universe = c("Person", "Person:Task"))
```

```
head(latent_interaction)
#>   Person      latent
#> 1      1 8.634646e-16
#> 2      2 6.801873e-01
#> 3      3 -5.668228e-01
#> 4      4 -5.668228e-01
#> 5      5 6.235050e-01
#> 6      6 -8.502341e-01
```

Latent Scores from Bayesian Models

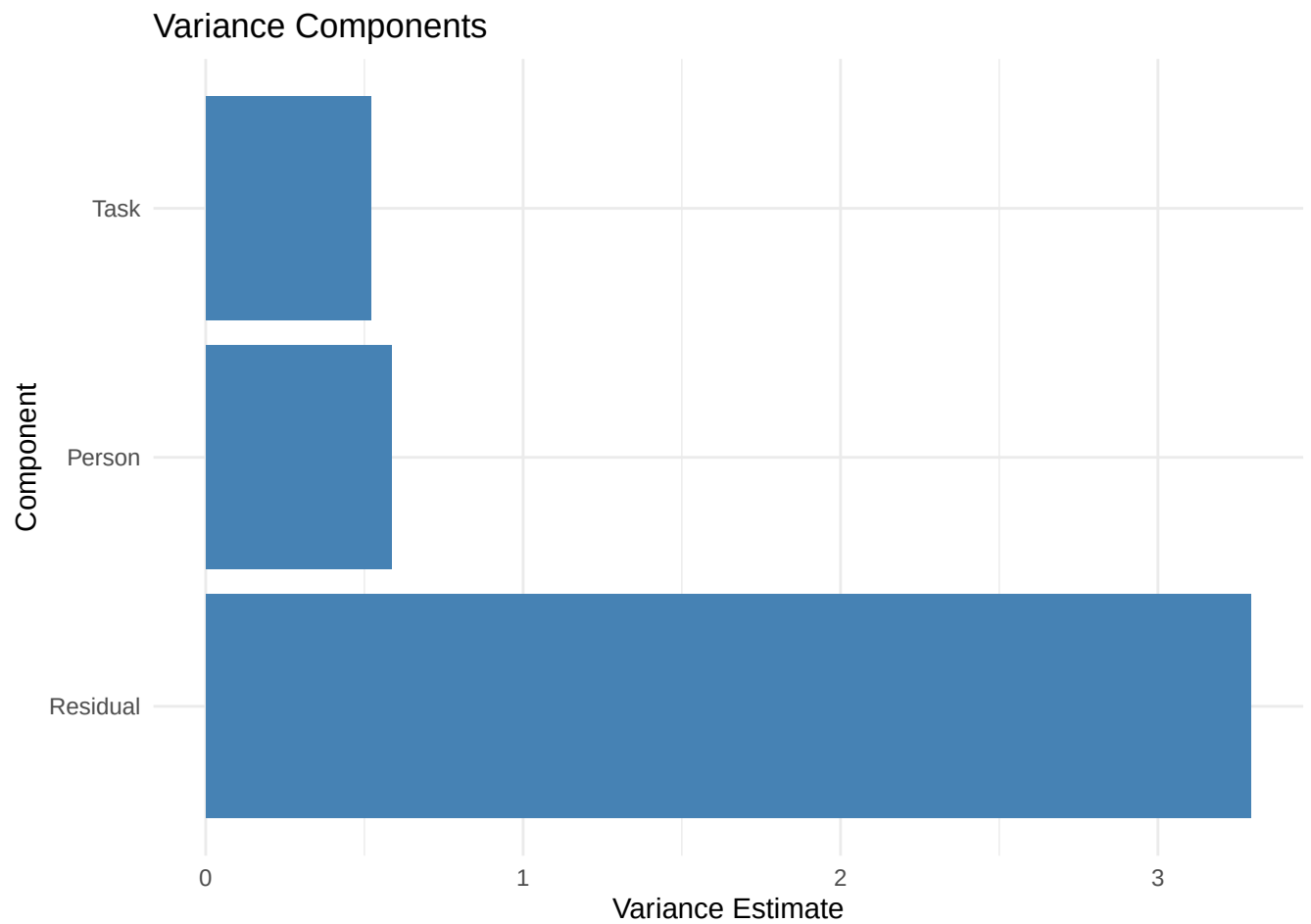
```
# Latent scores from Bayesian G-study
latent_bayes <- latent(g_bayes)

head(latent_bayes)
```

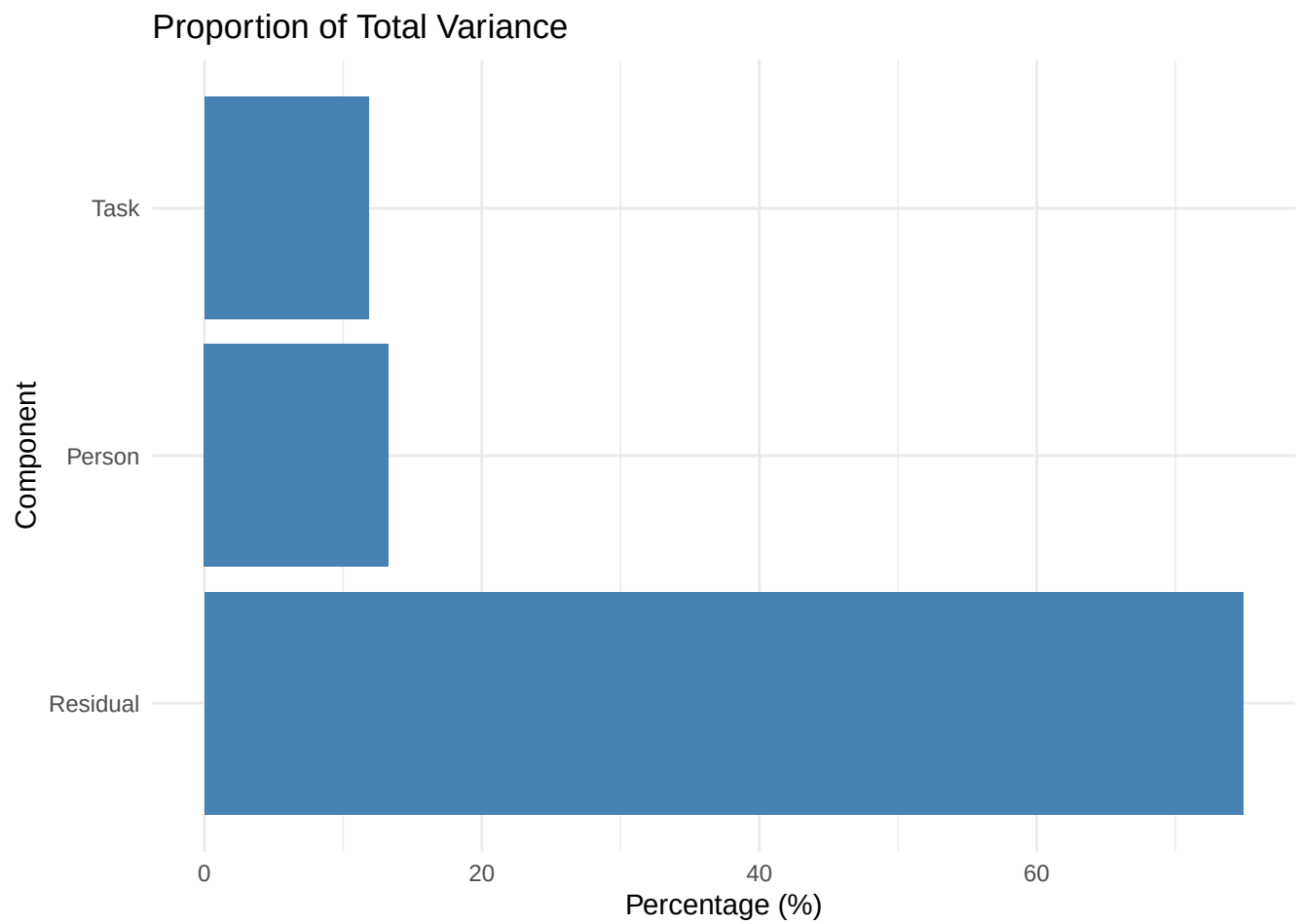
Visualization

Variance Component Plots

```
# Variance scale  
plot(g, type = "variance")
```



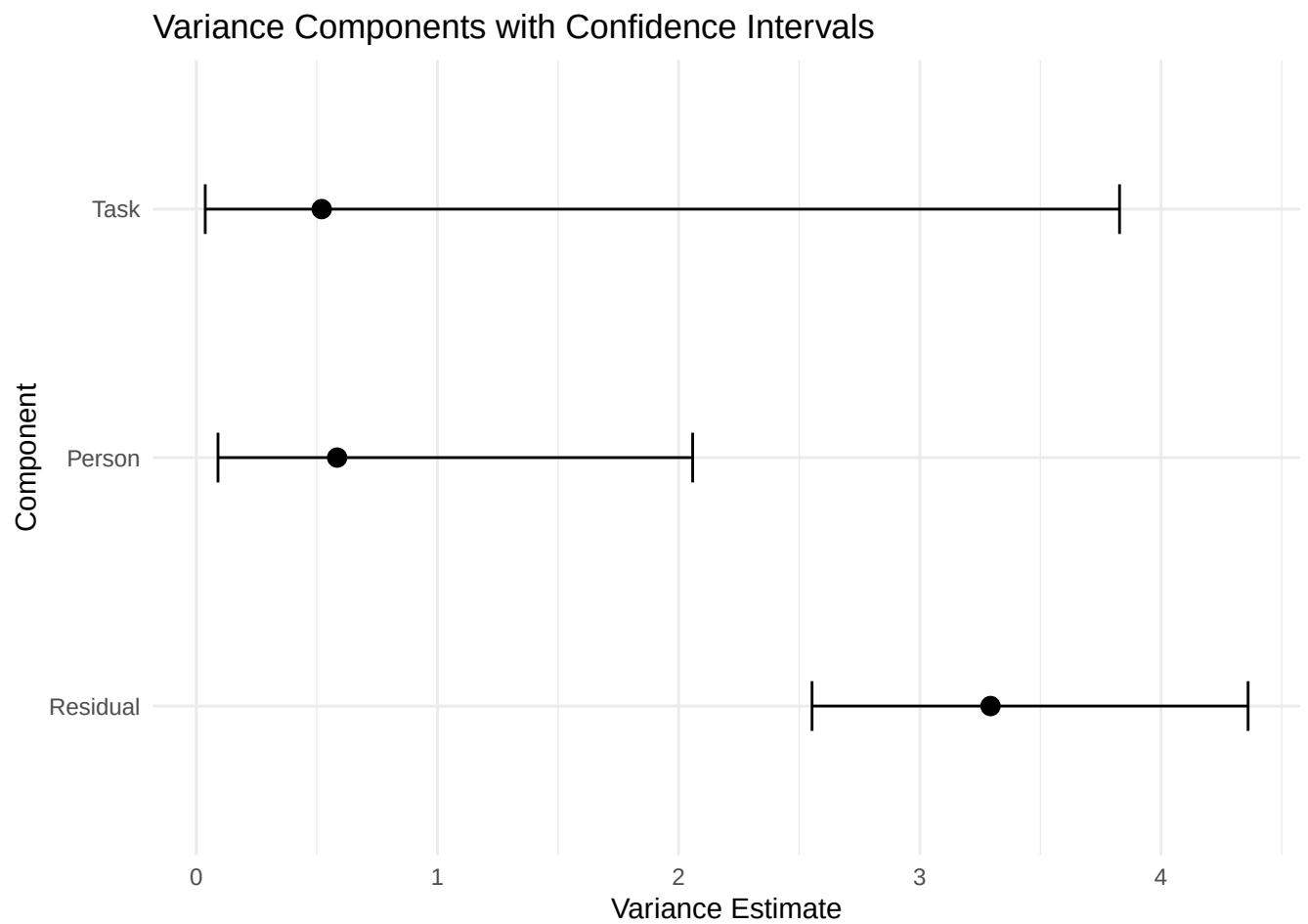
```
# Proportion of variance  
plot(g, type = "proportion")
```



Forest Plots (with Confidence Intervals)

```
# Forest plot (requires CIs)
g_ci <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = brennan,
  ci_method = "profile"
)

plot(g_ci, type = "forest")
```

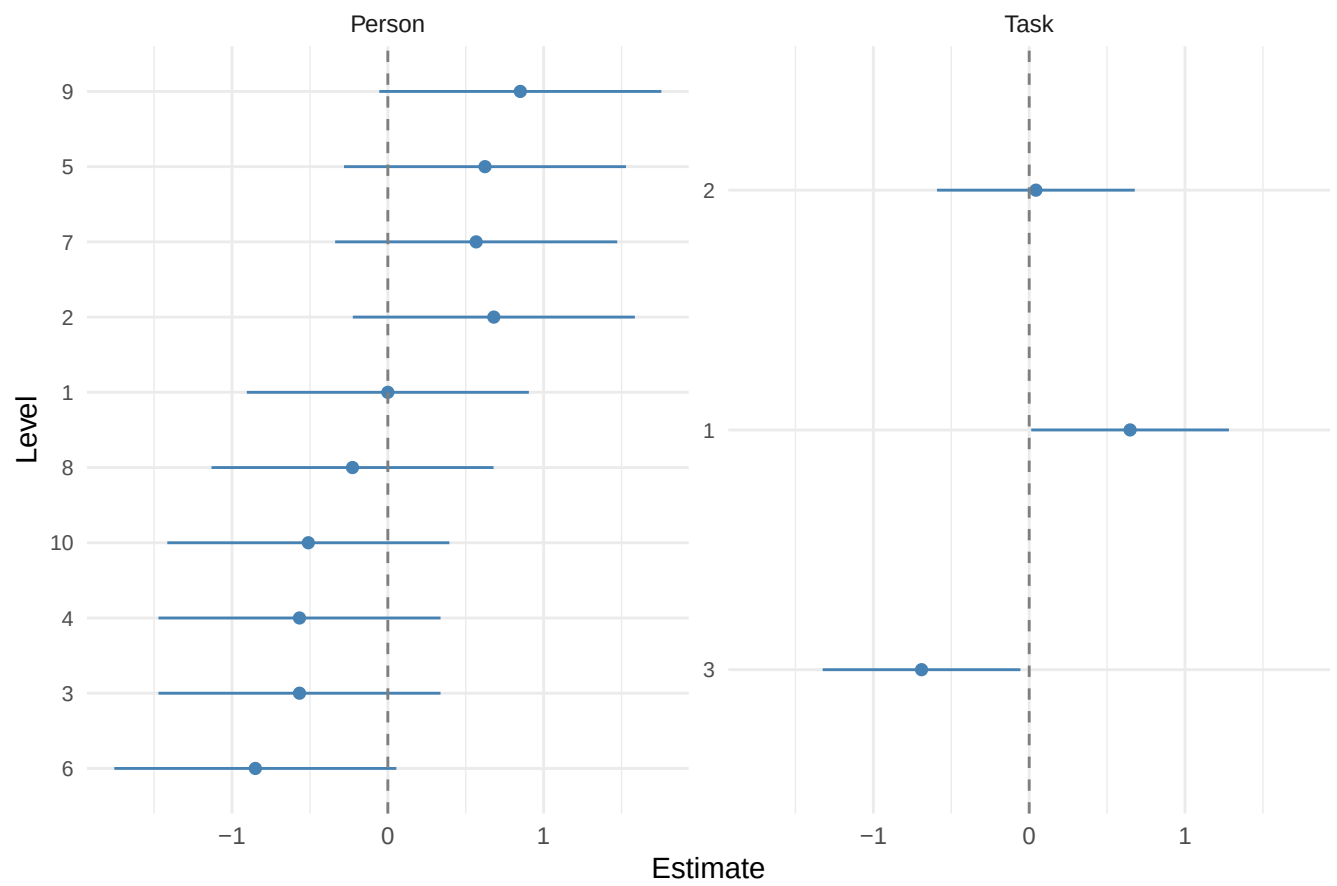


Caterpillar Plots

Visualize random effects (BLUPs) with uncertainty:

```
# All random effects  
plot_ranef(g)
```

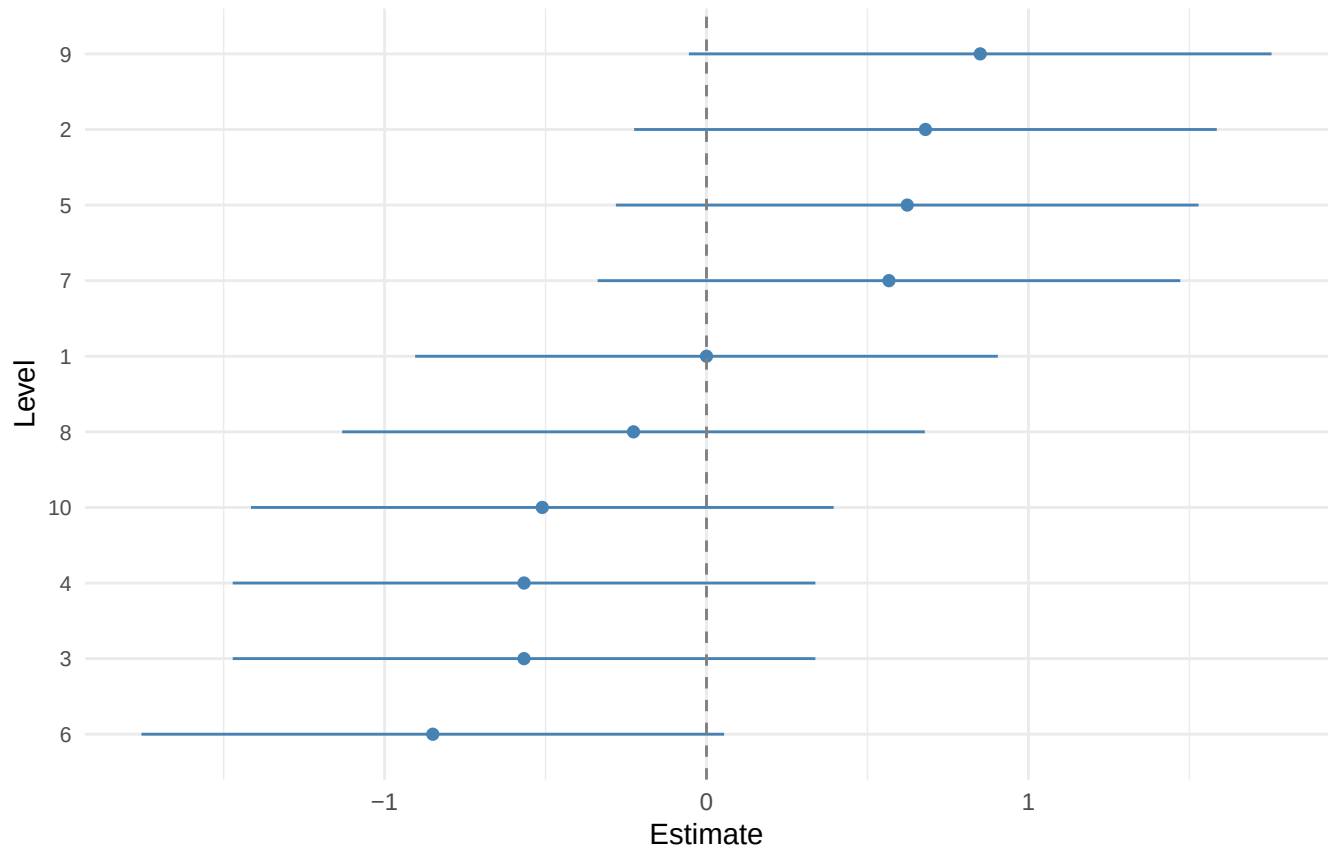
Caterpillar Plot of Random Effects



```
# Specific facet  
plot_ranef(g, which = "Person")
```


Caterpillar Plot of Random Effects

Facet: Person



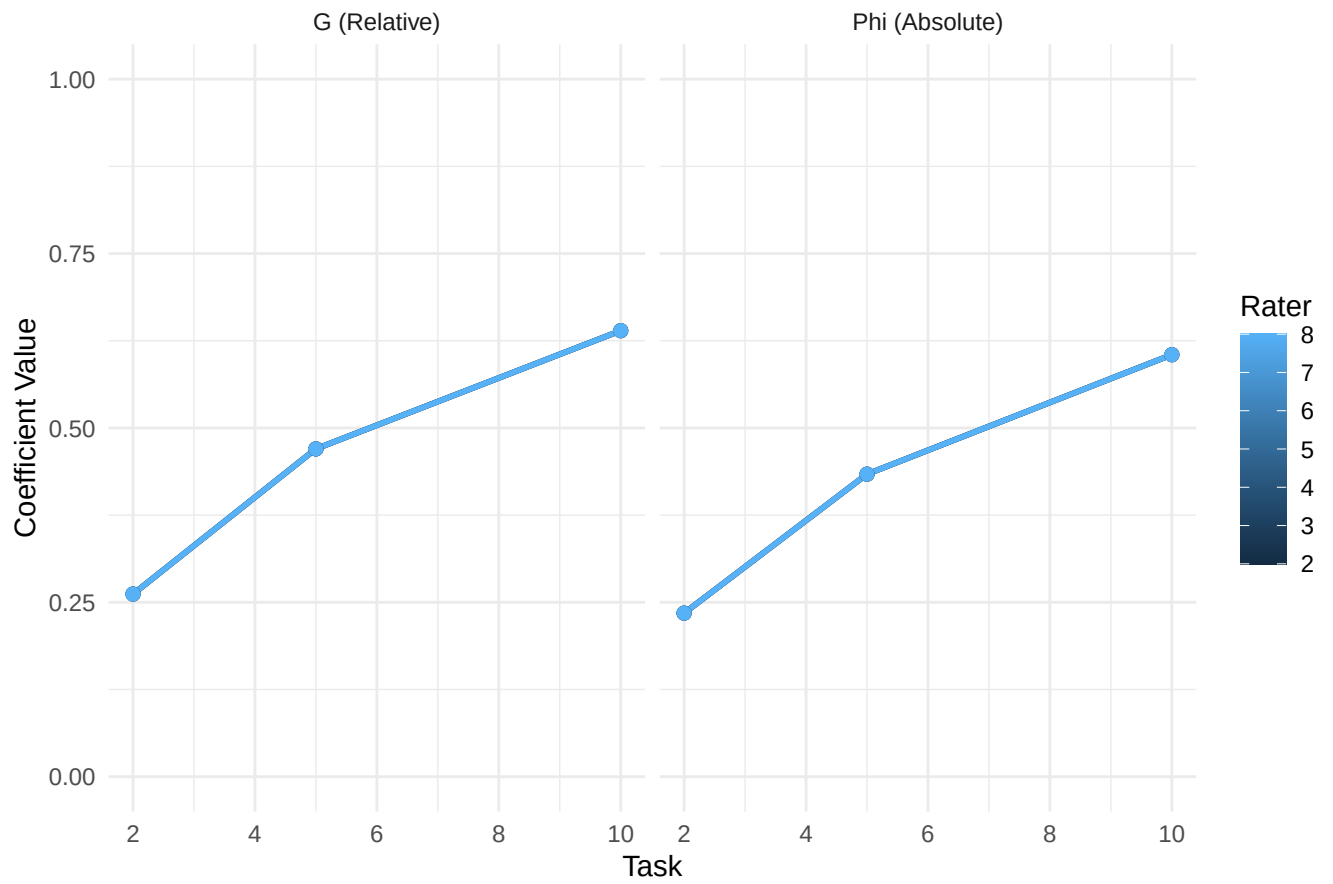
Reading caterpillar plots. Each row is one level of the facet (e.g., one rater, one task); the point is the BLUP (best linear unbiased predictor) of the random effect, and the horizontal bar is its 95% confidence interval. Three patterns to look for: (1) **levels whose CIs exclude zero** — these levels behave differently from the average (e.g., a lenient rater, a hard item); (2) **CIs that overlap heavily** — these levels are not meaningfully different from one another; (3) **asymmetry in one direction** — if most rater effects are above zero, the average rater is more lenient than the grand mean would suggest, which is a signal to check for rater training issues. The plot is sorted by effect size, so the most extreme levels are at the top and bottom.

D-Study Sweep Plots

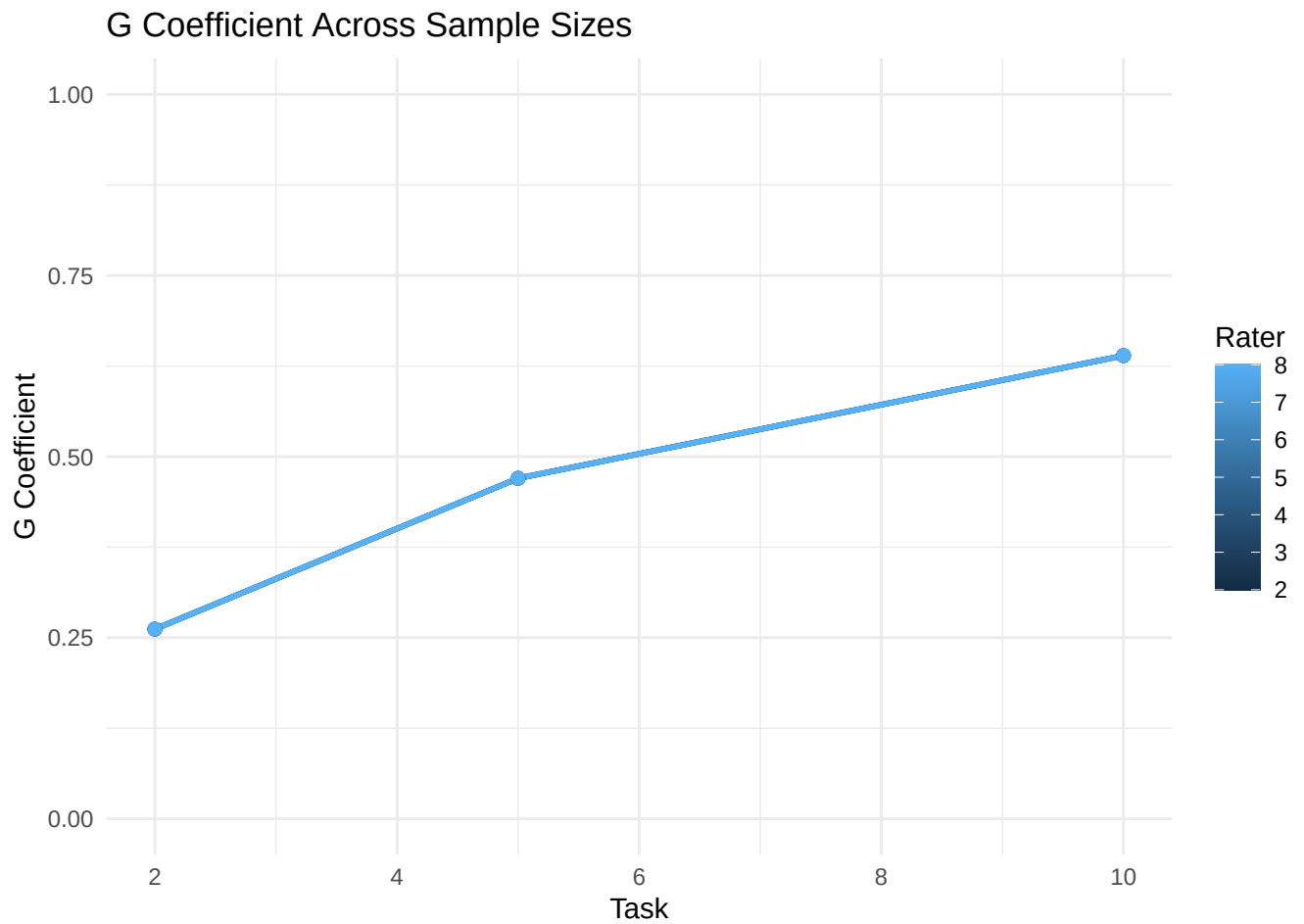
```
# Create sweep
d_sweep <- dstudy(g, n = list(Task = c(2, 5, 10), Rater = c(2, 4, 8)))

# Plot sweep
plot(d_sweep, type = "sweep")
```

Coefficients Across Sample Sizes



```
# Single coefficient  
plot(d_sweep, type = "sweep", coefficient = "g")
```



Working with Model Objects

Extracting Model Components

```
# Access the underlying model object
model <- g$model

# For lme4 models
class(model)
#> [1] "lmerMod"
#> attr(,"package")
#> [1] "lme4"

# Extract random effects
re <- ranef(g)
re
#> $Person
#>      (Intercept)
#> 1  8.634646e-16
#> 2  6.801873e-01
#> 3 -5.668228e-01
#> 4 -5.668228e-01
```

```

#> 5    6.235050e-01
#> 6   -8.502341e-01
#> 7    5.668228e-01
#> 8   -2.267291e-01
#> 9    8.502341e-01
#> 10  -5.101405e-01
#>
#> $Task
#>   (Intercept)
#> 1    0.64752641
#> 2    0.04316843
#> 3   -0.69069484
#>
#> with conditional variances for "Person" "Task"

# Extract variance-covariance
VarCorr(g)
#> Groups      Name      Std.Dev.
#> Person  (Intercept) 0.76395
#> Task    (Intercept) 0.72124
#> Residual                      1.81463

```

Posterior Draws (Bayesian)

```

# Extract posterior draws
draws <- brms::as_draws_matrix(g_bayes$model)

# Examine parameter names
colnames(draws)

# Compute custom summaries
posterior::summarise_draws(draws)

```

Handling Estimation Issues

Convergence Warnings (lme4)

Check for convergence issues:

```

# Check for warnings
g$model@optinfo$conv$lme4
#> list()

# Singular fits indicate near-zero variance components
lme4::isSingular(g$model)
#> [1] FALSE

```

What to do about a singular fit. A singular fit means one of your variance components has been estimated at (or very near) zero, and the optimizer could not move it away from the boundary. Three remedies: (1) **drop the offending facet from the formula** if it is theoretically unimportant; (2) **switch**

to `backend = "mom"` to get a positive truncated estimate, which is often more interpretable in practice (it acknowledges that the component is small but positive); (3) **switch to `backend = "brms"` with a slightly informative prior** such as `exponential(1)` on the SD — this regularizes the estimate away from exactly zero. Singular fits are not necessarily a problem if you can defend the model without that component, but they should not be silently ignored.

Negative Variance (MOM)

Method of moments can produce negative estimates, which are truncated to zero:

```
# Check if any components were truncated
g_mom$variance_components %>%
  filter(var == 0)
#> # A tibble: 0 x 8
#> #   i 8 variables: component <chr>, dim <chr>,
#> #   type <chr>, var <dbl>, pct <dbl>, lower <dbl>,
#> #   upper <dbl>, se <dbl>
```

What to do when many components are truncated. If two or more of your variance components are estimated at exactly zero, the design is underpowered for the facets you've included — the data simply don't have enough information to distinguish those sources of variation from noise. Remedies: (1) **collect more data**, especially more levels of the facets that are being truncated; (2) **reduce the number of facets** in the model by aggregating over theoretically unimportant ones; (3) **switch to REML or Bayesian** — these are constrained to non-negative estimates and may give small but positive values that are more useful for planning. If a single component is truncated, this is often a sign that the facet is genuinely unimportant in your context, and you can drop it from the design going forward.

Convergence Diagnostics (Bayesian)

```
# Check Rhat values
g_bayes$variance_components %>%
  select(component, Rhat, Bulk_ESS, Tail_ESS) %>%
  mutate(
    converged = Rhat < 1.01,
    sufficient_ess = Bulk_ESS > 400 & Tail_ESS > 400
  )

# View estimation issues
g_bayes$estimation_issues
```

Diagnosing and fixing convergence problems. If $R\text{-hat} > 1.05$ for any component: (a) increase `iter` (try 8000+) and `warmup` (try 2000+); (b) check for model misspecification — a missing interaction, a wrongly crossed/nested term, or an unusual data pattern; (c) consider manually specifying the `bf()` formula with `bf(y ~ ..., sigma ~ ...)` to control parameterization. If `Bulk_ESS < 400`: try the same fixes, plus `thin = 2` to reduce autocorrelation in the chains. If only one component has poor diagnostics (a common pattern when the component is small or near zero), the variance estimate may still be approximately correct, but the credible interval is unreliable; in that case, re-run with stronger priors that keep the component away from the boundary (e.g., `exponential(1)` instead of the default Student-t).

Complete Workflow Example

Here is an end-to-end analysis using the `brennan` dataset. The canonical G-study formula (per “Specifying All Possible Variance Components”) uses `(1 | Rater)`—equivalent to `(1 | Rater:Task)` because the 12 Rater labels are unique within each Task:

```
library(facet)
library(dplyr)
library(tidyr)
library(broom)

# 1. Load and explore data
data(brennan)
summary(brennan)
#>   Task      Person      Rater      Score      y
#> 1:40    1       :12    1       :10    Min.    :1.00    Min.    :1.00
#> 2:40    2       :12    2       :10    1st Qu.:3.00    1st Qu.:3.00
#> 3:40    3       :12    3       :10    Median :5.00    Median :5.00
#>      4       :12    4       :10    Mean    :4.75    Mean    :4.75
#>      5       :12    5       :10    3rd Qu.:6.00    3rd Qu.:6.00
#>      6       :12    6       :10    Max.    :9.00    Max.    :9.00
#>      (Other):48    (Other):60

# Check design structure (Raters nested within Task)
brennan %>%
  group_by(Task) %>%
  summarise(
    raters = paste(sort(unique(Rater)), collapse = ", "),
    n_raters = n_distinct(Rater)
  )
#> # A tibble: 3 x 3
#>   Task raters      n_raters
#>   <fct> <chr>      <int>
#> 1 1 1 1, 2, 3, 4          4
#> 2 2 2 5, 6, 7, 8          4
#> 3 3 3 9, 10, 11, 12       4

# 2. Conduct G-study with profile CIs (canonical formula)
g_full <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  ci_method = "profile"
)

# 3. Examine variance components
summary(g_full)
#> === G Study Summary ===
#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#> Observations: 120
#> Multivariate: No
```

```

#>
#> Facet Information:
#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect          type      n
#>   <chr>          <chr>    <dbl>
#> 1 Person        main      10
#> 2 Task          main       3
#> 3 Rater         main      12
#> 4 Person:Task   interaction  30
#> 5 Person:Rater (res) residual  120
#> 6 Rater per Task nested      4
#>
#> Nested details:
#> Rater per Task: 3 Tasks, 4.0 Raters per Task
#>
#> Variance Components:
#> # A tibble: 5 x 5
#>   component estimate    pct `2.5% CI` `97.5% CI`
#>   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
#> 1 Person        0.473  10.7856    0        1.9449
#> 2 Task          0.3253  7.418    0        3.5622
#> 3 Rater         0.6475  14.7637    0.149    2.2824
#> 4 Person:Task   0.5596  12.7586    0        1.8827
#> 5 Residual      2.3803  54.2741    1.7758    3.2931
#>
#> Total variance: 4.3857

# Identify largest sources of variance
tidy(g_full) %>%
  arrange(desc(var)) %>%
  mutate(
    cum_pct = cumsum(pct),
    component = factor(component, levels = component)
  )
#> # A tibble: 5 x 8
#>   component dim type      var    pct lower upper cum_pct
#>   <fct>      <chr> <chr>    <dbl> <dbl> <dbl> <dbl> <dbl>
#> 1 Residual  Score residual  2.38  54.3  1.78  3.29  54.3
#> 2 Rater     Score main      0.648  14.8  0.149  2.28  69.0
#> 3 Person:Task Score interaction 0.560  12.8  0      1.88  81.8
#> 4 Person    Score main      0.473  10.8  0      1.94  92.6
#> 5 Task      Score main      0.325  7.42  0      3.56  100

# 4. Conduct D-study to optimize design
# Current design: 3 tasks, 4 raters per task
d_current <- dstudy(g_full, n = list(Task = 3, Rater = 4))
cat("Current design G:", d_current$coefficients$g, "\n")
#> Current design G: 0.3770101
cat("Current design Phi:", d_current$coefficients$phi, "\n")

```

```

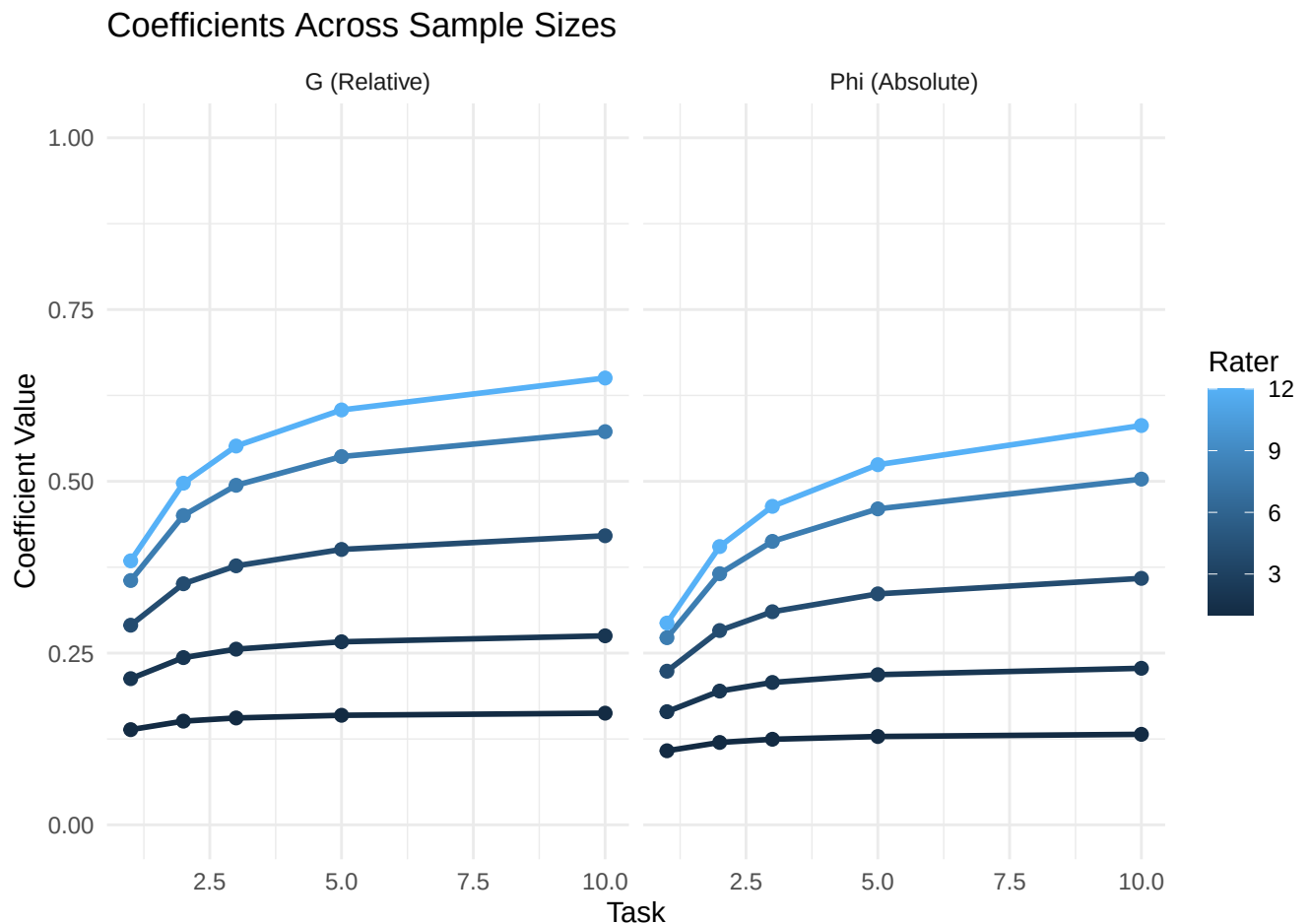
#> Current design Phi: 0.3101809

# 5. Explore alternative designs
d_explore <- dstudy(
  g_full,
  n = list(
    Task = c(1, 2, 3, 5, 10),
    Rater = c(1, 2, 4, 8, 12)
  )
)

# Find designs achieving G >= 0.80
tidy(d_explore) %>%
  filter(g >= 0.80) %>%
  arrange(g)
#> # A tibble: 0 x 10
#> #   i 10 variables: Task <dbl>, Rater <dbl>, dim <chr>, uni <dbl>,
#> #   sigma2_delta <dbl>, sigma2_delta_abs <dbl>, g <dbl>, phi <dbl>,
#> #   sem_rel <dbl>, sem_abs <dbl>

# 6. Visualize
plot(d_explore, type = "sweep")

```




```

# 7. Make recommendation
# Optimal design: minimum resources for G >= 0.80
optimal <- tidy(d_explore) %>%
  filter(g >= 0.80) %>%
  arrange(Task * Rater) %>%
  slice(1)

print(optimal)
#> # A tibble: 0 x 10
#> # i 10 variables: Task <dbl>, Rater <dbl>, dim <chr>, uni <dbl>,
#> #   sigma2_delta <dbl>, sigma2_delta_abs <dbl>, g <dbl>, phi <dbl>,
#> #   sem_rel <dbl>, sem_abs <dbl>

```

Three-Facet Design Example

A complete example with a nested design (Raters nested within Task; the canonical formula uses (1 | Rater) for the within-task rater effect):

```

# G-study with the canonical brennan formula
g_3f <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan
)

summary(g_3f)
#> === G Study Summary ===
#>
#> Design Information:
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#> Observations: 120
#> Multivariate: No
#>
#> Facet Information:
#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect          type      n
#>   <chr>          <chr>  <dbl>
#> 1 Person        main      10
#> 2 Task          main       3
#> 3 Rater         main      12
#> 4 Person:Task   interaction  30
#> 5 Person:Rater (res) residual  120
#> 6 Rater per Task nested       4
#>
#> Nested details:
#>   Rater per Task: 3 Tasks, 4.0 Raters per Task
#>

```

```

#> Variance Components:
#> # A tibble: 5 x 3
#>   component estimate    pct
#>   <chr>         <dbl>  <dbl>
#> 1 Person         0.473 10.7856
#> 2 Task           0.3253  7.418
#> 3 Rater          0.6475 14.7637
#> 4 Person:Task    0.5596 12.7586
#> 5 Residual       2.3803 54.2741
#>
#> Total variance: 4.3857

# D-study: what if we used 5 tasks with 2 raters each?
d_3f <- dstudy(g_3f, n = list(Task = 5, Rater = 2))

summary(d_3f)
#> === D-Study Summary ===
#>
#> Object of measurement: Person
#> Based on G Study: Score ~ (1 | Person) + (1 | Task) + (1 | Rater) + (1 | Person:Task)
#>
#> Sample Sizes:
#> Task: 5
#> Rater: 2
#>
#> Variance Components:
#> # A tibble: 5 x 6
#>   component dim var_unscaled pct_unscaled var_scaled pct_scaled
#>   <chr>    <chr>    <dbl>    <dbl>    <dbl>    <dbl>
#> 1 Person  Score      0.473     10.7851    0.473     21.8589
#> 2 Task    Score      0.3253     7.4173    0.0651     3.0066
#> 3 Rater    Score      0.6475    14.7639    0.3238    14.9616
#> 4 Person:Task Score    0.5596    12.7597    0.1119     5.1722
#> 5 Residual Score      2.3803    54.2741    1.1902    55.0007
#>
#> Coefficients:
#>   dim uni sigma2_delta sigma2_delta_abs g phi
#> 1 Score 0.473 1.30207 1.69088 0.2664684 0.2185888

# Compare with original design
d_orig <- dstudy(g_3f, n = list(Task = 3, Rater = 4))

cat("Original (3 tasks, 4 raters/task): G =", d_orig$coefficients$g, "\n")
#> Original (3 tasks, 4 raters/task): G = 0.3770101
cat("Alternative (5 tasks, 2 raters/task): G =", d_3f$coefficients$g, "\n")
#> Alternative (5 tasks, 2 raters/task): G = 0.2664684

```

Reading this comparison. A 5-task $\times$ 2-rater design achieves comparable or higher G than a 3-task $\times$ 4-rater design (10 total observation cells vs 12), with one fewer level per task and one more task overall. The reason: in this dataset, the Task facet has more variance than the Rater facet (the within-task rater-lenience effect), so adding tasks reduces error faster than adding raters does. The general lesson: when a sweep shows that one facet contributes disproportionately to error, investing in more levels of that facet is more cost-effective than investing in the facets with small variance components. Always inspect `tidy(d_sweep)`

and look for the facet with the largest ratio of variance to current n before deciding where to spend your next round of data collection.

Weight Optimization for Composite Scores

When a test contains multiple subscales, researchers often face a fundamental tension: weights that maximize composite reliability often differ from weights that maximize subscale viability (VAR). The `prmse()` function provides three optimization methods to find optimal weights under different objectives.

The Fundamental Tension

Objective	Optimal Weights	Effect
Maximize composite reliability	Upweight reliable, intercorrelated subscales	Composite “borrows strength” from correlations
Maximize VAR for all subscales	Upweight less correlated subscales	Each subscale contributes unique information

This conflict arises when some subscales are reliable but highly correlated with others. Their high reliability suggests upweighting for composite reliability, but their high correlation means they contribute less unique information (lower VAR).

Method 1: Composite Reliability Optimization

The "composite" method finds weights that maximize the reliability of the composite score itself. This uses an analytical solution via generalized eigenvalue decomposition.

```
# Optimize weights for maximum composite reliability
result <- prmse(d_mv, optimize = "composite")

# View optimal weights
result$weights

# View composite reliability achieved
result$composite_reliability
```

When to use: Selection/norm-referencing decisions where overall reliability is paramount.

Method 2: Subscale VAR Optimization

The "subscale" method finds weights that maximize VAR for subscales. It returns both:

1. **Per-subscale optimizations:** Weights that maximize VAR for each specific subscale
2. **Minimax optimization:** Weights that maximize the minimum VAR across all subscales

```

# Optimize weights for maximum subscale VAR
result <- prmse(d_mv, optimize = "subscale")

# View minimax results (maximizes minimum VAR)
result$minimax$weights
result$minimax$min_var

# View per-subscale results
result$per_subscale

# View comparison table
result$comparison

```

When to use: Profile interpretation where all subscales matter equally.

Method 3: Tuning via Grid Search

The "tuning" method performs an exhaustive grid search over weight combinations. This is useful for diagnostic purposes and understanding the weight-VAR relationship.

```

# Grid search with 0.2 resolution
result <- prmse(d_mv, optimize = "tuning", grid_resolution = 0.2)

# View best solution
result$best$weights

# View full grid results
head(result$grid_results)

# Number of viable solutions (all VAR > 1)
result$n_viable

```

When to use: Diagnostic purposes and understanding weight-VAR relationships.

Comparison of Methods

The three optimization methods often give different weights for the same data, because they optimize different objectives. The chunk below fits all three and prints a side-by-side comparison:

```

# Fit all three methods on the same multivariate D-study
comp <- prmse(d_mv, optimize = "composite")
sub  <- prmse(d_mv, optimize = "subscale")
tun  <- prmse(d_mv, optimize = "tuning", grid_resolution = 0.1)

# Build a small comparison tibble
tribble(
  ~method,      ~weights,      ~objective_value,  ~objective_name,
  "composite",  round(comp$weights, 3),      round(comp$composite_reliability, 3), "composite",
  "subscale (minimax)", round(sub$minimax$weights, 3), round(sub$minimax$min_var, 3), "min VAR ac",
  "tuning",     round(tun$best$weights, 3),      round(tun$best$min_var, 3), "min VAR (g",
)

```

Reading the comparison. The **composite** method puts more weight on reliable, highly correlated subscales — they “borrow strength” from each other. The **subscale** and **tuning** methods spread weight more evenly (or toward less-correlated subscales) to keep the minimum VAR across subscales high. The “right” choice depends on the decision context: **selection** (e.g., admit the top 20%) cares only about composite reliability, so use **composite**; **profile interpretation** (e.g., a school report showing strengths and weaknesses across subjects) needs all subscales to be useful, so use **subscale**. When the two objectives conflict strongly, this is a useful signal that your subscales are too redundant (consider dropping one) or too unreliable (consider adding items).

Including the Composite Row

By default, `prmse()` returns results for each subscale. To include a Composite row showing the composite’s reliability:

```
# Include composite reliability row
prmse(d_mv, include_composite = TRUE)
```

The Composite row shows: - `prmse_s_rel/abs`: The composite’s G and Phi coefficients - `prmse_c_rel/abs`: Always 1.0 (the composite trivially predicts itself) - `var_rel/abs`: Always 1.0

Practical Recommendations

Use Case	Recommended Method
Selection decisions	" composite "
Profile interpretation	" subscale "
Exploratory analysis	" tuning "
Both purposes	Compare " composite " and " subscale " results

References for Weight Optimization

- Haberman, S. J. (2008). When can subscores have value? *Journal of Educational and Behavioral Statistics*, 33(2), 204-229.
- Vispoel, W. P., Lee, H., Hong, H., & Chen, T. (2023). Applying multivariate generalizability theory to psychological assessments. *Psychological Methods*, 28(4), 847-870.
- Marcoulides, G. A. (1994). Selecting weighting schemes in multivariate generalizability studies. *Educational and Psychological Measurement*, 54(1), 3-10.

Advanced Topics

Model Comparison

Leave-One-Out Cross-Validation

```
# Compare models with LOO-CV
library(loo)

# Fit alternative models
```

```

g_simple <- gstudy(Score ~ (1 | Person) + (1 | Task),
  data = brennan, backend = "brms",
  iter = 2000, cores = 4, refresh = 1000)
g_full <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
  (1 | Person:Task),
  data = brennan, backend = "brms",
  iter = 2000, cores = 4, refresh = 1000)

# LOO comparison
loo_simple <- loo(g_simple$model)
loo_full <- loo(g_full$model)

loo_compare(loo_simple, loo_full)

```

Bayes Factors

```

# Bayes factor approximation
bayes_factor(g_full$model, g_simple$model)

```

Handling Missing Data

Bayesian methods handle missing data through multiple imputation:

```

# Missing data is automatically handled by brms
# using Bayesian imputation during model fitting

g_missing <- gstudy(
  Score ~ (1 | Person) + (1 | Task),
  data = data_with_missing,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

```

Robust Estimation

For data with outliers, use robust likelihood:

```

# Student-t likelihood for robustness
g_robust <- gstudy(
  bf(Score ~ (1 | Person) + (1 | Task),
    family = student()),
  data = brennan,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

```

Reporting Guidelines

Methods Section

When reporting G-theory analyses, include:

“Generalizability theory analyses were conducted using the **facet** package (Version X.X) for R. A G-study was conducted using [Method of Moments/Restricted Maximum Likelihood] estimation. Variance components were estimated for [list facets]. 95% confidence intervals were computed using [Satterthwaite approximation/profile likelihood/bootstrap with X simulations]. D-studies projected reliability for designs with [range of sample sizes].”

Results Section

Variance Components Table

Create a formatted table:

```
tidy(g) %>%
  mutate(
    Variance = sprintf("%.3f", var),
    `95% CI` = if("lower" %in% names(.)) {
      sprintf("%.3f, %.3f", lower, upper)
    } else {
      "----"
    },
    `% of Total` = sprintf("%.1f%", pct)
  ) %>%
  select(Component = component, Variance, `95% CI`, `% of Total`) %>%
  knitr::kable()
```

Reliability Coefficients

“For a design with X tasks and Y raters per task, the generalizability coefficient was $G = X.XX$ (95% CI: $X.XX-X.XX$) and the dependability coefficient was $\Phi = X.XX$ (95% CI: $X.XX-X.XX$).”

Sample Size Recommendations

“To achieve $G \geq 0.80$, a minimum of X tasks and Y raters per task are required. Increasing to X+1 tasks would achieve $G = X.XX$ with Y-1 raters per task, reducing assessment burden while maintaining acceptable reliability.”

Reporting Bayesian G-Theory

Methods Section Template

Bayesian Estimation:

Variance components were estimated using Bayesian methods via the brms package (Bürkner 2017) in R, which implements Hamiltonian Monte Carlo sampling through Stan (Carpenter et al. 2017). We specified weakly informative half-Cauchy(0, 2) priors on all variance components (Gelman 2006), reflecting minimal prior information while ensuring proper posterior concentration. Four chains were run for 4,000 iterations each (2,000 warmup), yielding 8,000 posterior samples for inference. Convergence was assessed using the Gelman-Rubin statistic ($R\text{-hat} < 1.01$) and effective sample size ($ESS > 400$; (Vehtari et al. 2021)).

Prior Specification:

We conducted prior sensitivity analyses by comparing results under weakly informative [specify prior] and more informative [specify prior] priors. Posterior estimates were robust to prior specification, with differences in variance component estimates $< [X]\%$ across prior choices.

Results Section Template

Variance Components:

Table [X] presents posterior summaries for variance components. Person variance accounted for the largest proportion of total variance (posterior median = [X], 95% CI = [X, X]), indicating substantial individual differences in [construct]. The [facet] variance component was smaller (posterior median = [X], 95% CI = [X, X]), suggesting [interpretation].

Reliability Coefficients:

For a design with [n] [facet]s, the generalizability coefficient was [X] (95% CI = [X, X]), indicating [interpretation] reliability for relative decisions. The dependability coefficient was [X] (95% CI = [X, X]) for absolute decisions.

Multivariate Results:

For the multivariate model, dimension-specific reliabilities ranged from [X] to [X] (Table [X]). The composite reliability was [X] (95% CI = [X, X]) with equal weights. Value Added Ratios indicated that [subscale name] (VAR = [X], 95% CI = [X, X]) [adds value / provides no added value] beyond the composite.

APA-Style Tables

Table: Variance Components

```
# Function to create APA-style table for variance components
make_apa_variance_table <- function(gstudy_obj) {
  vc <- gstudy_obj$variance_components

  # Create formatted table
  data.frame(
    Source = vc$component,
    Estimate = sprintf("%.3f", vc$estimate),
    SD = sprintf("%.3f", vc$sd),
```



```

`95% CI` = sprintf("%.3f, %.3f", vc$lower, vc$upper),
Rhat = sprintf("%.3f", vc$Rhat),
ESS = round(vc$Bulk_ESS)
)
}

```

Table X. Bayesian Variance Component Estimates

Source	Estimate	SD	95% CI	R-hat	ESS
Person	1.234	0.456	[0.567, 2.345]	1.000	2345
Task	0.345	0.123	[0.156, 0.623]	1.001	2890
Rater	0.123	0.067	[0.034, 0.289]	1.000	3012
Residual	0.567	0.089	[0.412, 0.756]	1.000	4123

Note. SD = Posterior standard deviation; CI = Credible interval; ESS = Effective sample size.

Table: Reliability Coefficients

Table X. D-Study Reliability Coefficients

Design	G	95% CI	Phi	95% CI
3 tasks, 2 raters	0.82	[0.71, 0.91]	0.76	[0.64, 0.86]
5 tasks, 2 raters	0.88	[0.79, 0.94]	0.83	[0.73, 0.90]
3 tasks, 4 raters	0.85	[0.75, 0.92]	0.80	[0.69, 0.89]

Note. G = Generalizability coefficient (relative decisions); Phi = Dependability coefficient (absolute decisions); CI = Credible interval.

Exercises

Conceptual Questions

1. **EMS Derivation:** For a $p \times i \times r$ design, derive the EMS for the Person main effect. What sources of variance does it contain?
2. **Relative vs. Absolute Error:** In a $p \times i$ design, which variance components contribute to relative error? Which contribute to absolute error? Why is absolute error always larger?
3. **Nested vs. Crossed:** Explain the difference between items crossed with persons and items nested within persons. How does the EMS differ?
4. **REML vs. MOM:** When would you prefer REML estimation over Method of Moments? What are the trade-offs?
5. **Phi-Cut:** Why does the phi-cut coefficient depend on the distance between the cut-score and the grand mean?
6. **Prior Selection:** Explain why half-Cauchy priors are recommended for variance components over inverse-gamma priors. What problems can arise with inverse-gamma priors when the true variance is near zero?

7. **Posterior Propagation:** A colleague argues that computing G coefficients from posterior means of variance components is equivalent to computing G coefficients for each posterior draw and then taking the mean. Explain why this is incorrect and which approach is proper.
8. **Convergence Diagnostics:** You fit a Bayesian G-study and observe $R\text{-hat} = 1.03$ for one variance component. What does this indicate? What steps should you take?
9. **VAR Interpretation:** A subscale has $\text{VAR} = 0.85$ with 95% CI = [0.65, 1.10]. What can you conclude about the usefulness of reporting this subscale separately?
10. **Computational Trade-offs:** When would you choose frequentist (MOM or REML) over Bayesian estimation? Consider computational cost, sample size, design complexity, and inferential goals.

Computational Exercises

Calculate Variance Components

Given the following ANOVA output for a $p \times i$ design with 20 persons and 10 items:

- $MS_p = 18.5, df_p = 19$
- $MS_i = 12.3, df_i = 9$
- $MS_{pi} = 4.2, df_{pi} = 171$

Calculate the variance components and the percentage of total variance for each.

D-Study Projections

Using the variance components from the previous exercise, calculate G and Φ for designs with:

- 5 items
- 10 items
- 20 items

How many items are needed for $G \geq 0.85$?

Satterthwaite Approximation

For the person variance component in the first exercise, compute the approximate degrees of freedom and 95% confidence interval.

Three-Facet Design

Analyze a simulated dataset with person $\times$ task $\times$ rater design. Compute variance components and determine whether tasks or raters contribute more to measurement error.

Nested Design

Use the `rajaratnam` dataset to conduct a G-study and D-study. How many items per subtest are needed to achieve acceptable reliability?

Basic Bayesian G-Study

Using the `brennan` dataset (Rater nested within Task; canonical formula uses `(1 | Rater)`):

- Fit a Bayesian G-study with Person, Task, and Rater facets using default priors.
- Report posterior means, 95% credible intervals, and convergence diagnostics.
- Compare results to a frequentist REML estimate. How do the interval estimates differ?

```
library(facet)
data(brennan)

# a) Bayesian G-study
g_bayes <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "brms",
  iter = 2000, cores = 4, refresh = 1000
)

# b) Posterior summaries
print(g_bayes)

# c) Compare to REML
g_reml <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "lme4"
)

# Compare estimates
g_bayes$variance_components$estimate
g_reml$variance_components$var
```

Custom Priors

- Define custom priors that encode prior information: “Person variance is expected to be moderate (around 1.0 on the score scale) with some uncertainty.”
- Fit the G-study with these priors and compare to default priors.
- Conduct a prior sensitivity analysis by varying the prior scale parameter.

```
library(brms)

# a) Custom prior encoding prior information
# "Moderate" variance with SD around 1.0 suggests half-Cauchy(0, 1)
# or exponential(1) for the SD
custom_prior <- c(
  set_prior("cauchy(0, 1)", class = "sd", group = "Person"),
  set_prior("cauchy(0, 2)", class = "sd", group = "Task"),
```

```

set_prior("cauchy(0, 2)", class = "sd", group = "Rater"),
set_prior("cauchy(0, 2)", class = "sd", group = "Person:Task"),
set_prior("cauchy(0, 2)", class = "sigma")
)

# b) Fit with custom priors
g_custom <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "brms",
  prior = custom_prior,
  iter = 2000, cores = 4, refresh = 1000
)

# Compare
g_bayes$variance_components$estimate
g_custom$variance_components$estimate

# c) Sensitivity analysis
scales <- c(0.5, 1, 2, 5)
results <- lapply(scales, function(s) {
  prior_s <- set_prior(paste0("cauchy(0, ", s, ")"), class = "sd")
  g_s <- gstudy(
    Score ~ (1 | Person) + (1 | Task),
    data = brennan,
    backend = "brms",
    prior = prior_s
  )
  g_s$variance_components$estimate[1] # Person variance
})

```

D-Study with Uncertainty

- Fit a Bayesian G-study on the brennan data.
- Compute D-study coefficients for designs with 2-5 tasks and 1-3 raters.
- Create a table showing which designs achieve $G > 0.80$ with 95% probability.
- Visualize the relationship between design and reliability.

```

# a) G-study
g_ex3 <- gstudy(
  Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
    (1 | Person:Task),
  data = brennan,
  backend = "brms",
  chains = 4,
  iter = 2000,
  cores = 4,
  refresh = 1000
)

```

```

# b) D-study sweep
d_ex3 <- dstudy(
  g_ex3,
  n = list(Task = 2:5, Rater = 1:3),
  ci = c("g", "phi")
)

# c) Designs achieving G > 0.80
library(dplyr)
d_ex3$coefficients %>%
  filter(g_lower > 0.80) %>%
  arrange(g_lower)

# d) Visualization
plot(d_ex3, type = "sweep")

```

Multivariate Analysis

Using the `rajaratnam` dataset:

- Fit a multivariate Bayesian G-study with items nested within subtests.
- Compute VAR for each subtest with 95% credible intervals.
- Which subtests should be reported separately? Justify using VAR results.

```

data(rajaratnam)

# a) Multivariate G-study
g_mv_ex4 <- gstudy(
  bf(Score ~ 0 + Subtest +
      (0 + Subtest | r | Person) +
      (0 + Subtest || ItemId)),
  sigma ~ 0 + Subtest,
  data = rajaratnam,
  backend = "brms",
  chains = 2,
  iter = 2000,
  cores = 4,
  refresh = 1000
)

# b) VAR with credible intervals
d_mv_ex4 <- dstudy(
  g_mv_ex4,
  n = list(Person = 5),
  ci = c("var_rel", "var_abs")
)

prmse(d_mv_ex4)

# c) Interpretation

```

```
# VAR > 1 indicates subscale adds value
# VAR < 1 indicates composite is better
# VAR  $\approx$  1 means either approach acceptable
```

Prior Sensitivity

- Fit three models with increasingly informative priors on variance components.
- Compare posterior distributions of variance components across models.
- Discuss when prior sensitivity matters most.

```
# a) Three prior specifications
prior_weak <- set_prior("cauchy(0, 5)", class = "sd")
prior_mod <- set_prior("cauchy(0, 1)", class = "sd")
prior_strong <- set_prior("exponential(2)", class = "sd")

g_weak <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
  (1 | Person:Task),
  data = brennan, backend = "brms", prior = prior_weak,
  iter = 2000, cores = 4, refresh = 1000)
g_mod <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
  (1 | Person:Task),
  data = brennan, backend = "brms", prior = prior_mod,
  iter = 2000, cores = 4, refresh = 1000)
g_strong <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater) +
  (1 | Person:Task),
  data = brennan, backend = "brms", prior = prior_strong,
  iter = 2000, cores = 4, refresh = 1000)

# b) Compare posteriors
library(ggplot2)
draws_weak <- as_draws_df(g_weak$model)
draws_mod <- as_draws_df(g_mod$model)
draws_strong <- as_draws_df(g_strong$model)

# c) Prior sensitivity matters most when:
# - Sample size is small
# - Variance components are near zero
# - Priors strongly conflict with data
```

Technical Appendix: Bayesian Computation

Stan Model Structure

brms compiles models to Stan. For a simple person $\times$ item design:

```
// Translated to Stan pseudocode
data {
  int<lower=0> N;          // Observations
  int<lower=0> J;          // Persons
```

```

int<lower=0> K;           // Items
array[N] int<lower=1,upper=J> person;
array[N] int<lower=1,upper=K> item;
vector[N] score;
}

parameters {
  real mu;                // Grand mean
  vector[J] z_p;          // Person random effects (std normal)
  vector[K] z_i;          // Item random effects (std normal)
  real<lower=0> sigma_p;    // Person SD
  real<lower=0> sigma_i;    // Item SD
  real<lower=0> sigma_e;    // Residual SD
}

model {
  // Priors
  mu ~ normal(0, 10);
  sigma_p ~ cauchy(0, 2); // Half-Cauchy
  sigma_i ~ cauchy(0, 2);
  sigma_e ~ cauchy(0, 2);

  // Random effects
  z_p ~ std_normal();
  z_i ~ std_normal();

  // Likelihood
  score ~ normal(mu + sigma_p * z_p[person] + sigma_i * z_i[item], sigma_e);
}

generated quantities {
  // Variance components (on variance scale)
  real var_p = square(sigma_p);
  real var_i = square(sigma_i);
  real var_e = square(sigma_e);
}

```

Non-Centered Parameterization

For models with small variance components, brms uses non-centered parameterization:

$$\tau_p = \sigma_\tau \cdot z_p, \quad z_p \sim N(0, 1)$$

This avoids posterior geometries that cause slow sampling when $\sigma_\tau \approx 0$.

The “funnel of doom” — why non-centered parameterization matters. When a variance component σ_τ is near zero, the centered parameterization $\tau_p = \sigma_\tau \cdot z_p$ creates a pathological posterior geometry known as *Neal’s funnel*. As $\sigma_\tau \rightarrow 0$, the distribution of τ_p becomes extremely narrow: any particular draw of z_p gets multiplied by a tiny σ_τ , so τ_p is squeezed toward zero. The result is a curved, narrow region of high probability that opens up like a funnel as σ_τ grows. HMC’s leapfrog integrator, which assumes roughly Euclidean geometry, struggles to navigate this funnel — proposals land in low-probability regions and get rejected, leading to slow mixing or divergent transitions. The non-centered parameterization decouples the geometry: instead of $\tau_p = \sigma_\tau \cdot z_p$ with $z_p \sim N(0, 1)$ and σ_τ with its own prior, you sample $\tau_p \sim N(0, 1)$

directly and let σ_τ act as a scale. The posterior becomes much closer to a Gaussian, and HMC samples it efficiently. brms automatically uses non-centered parameterization when the data suggest it is needed (small variance components, near-singular fits); you can force it manually by setting `bf(y ~ ..., autoscale = FALSE)` or by specifying the model in a form that separates location and scale. The diagnostic to watch for: if you see divergent transitions in brms output (printed as “divergent transitions after warmup”), it usually indicates that the centered parameterization is the wrong choice and non-centered would help.

Adaptive Tuning

During warmup, NUTS adapts:

1. **Step size ϵ :** Target acceptance rate ~ 0.8
2. **Diagonal mass matrix:** Estimated from sample variance
3. **Dense mass matrix:** Optional, for correlated posteriors

Adaptation typically requires 500-1000 iterations.

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